

Chapter 4

Function overview

4.1 Position control mode

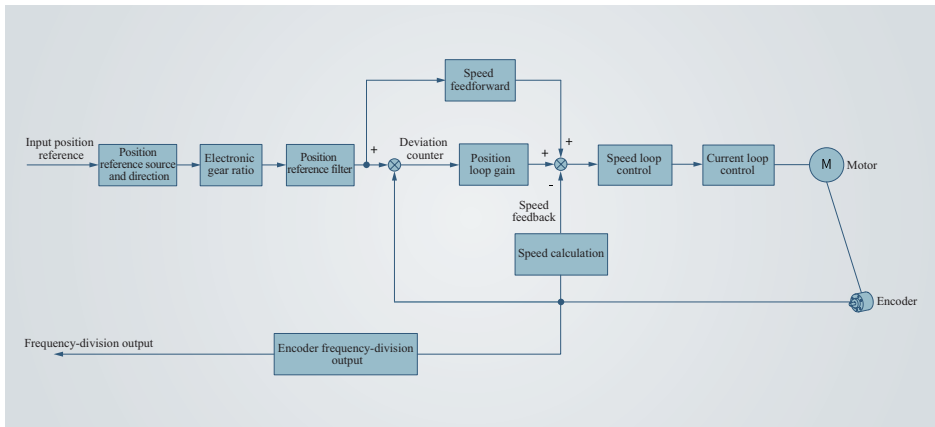


Figure 4-1 Position control block diagram

Set the parameter C00.00 = 0 through the servo drive panel or the drive debugging platform, and then the servo drive will work in the position control mode.

4.1.1 Block diagram of function codes in position control mode

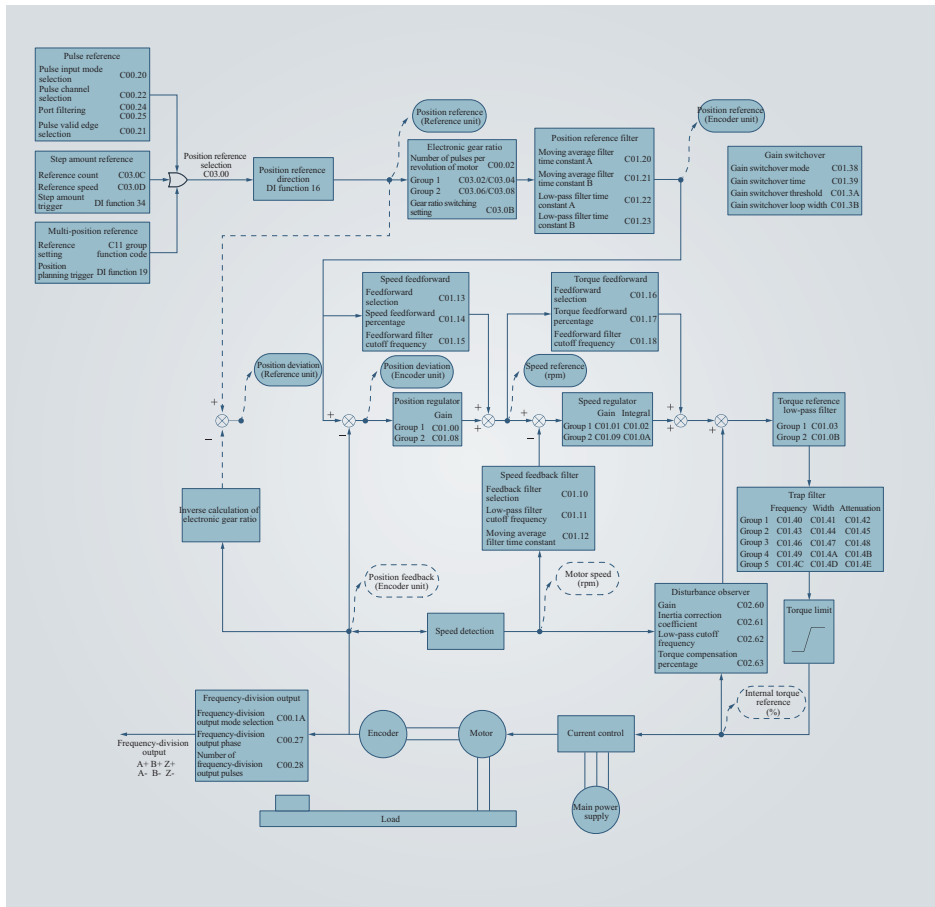


Figure 4-2 Block diagram of function codes in position control mode

4.1.2 Position reference input

Position reference inputs include: position reference source, position reference direction and position reference prohibition.

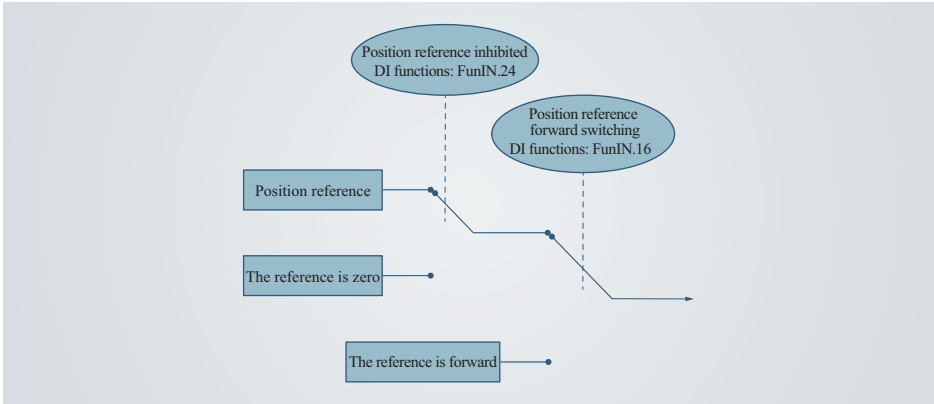


Figure 4-3 Block diagram of the position reference input setting

Position reference source

In position control mode, the position reference source should be set by parameter C03.00 first.

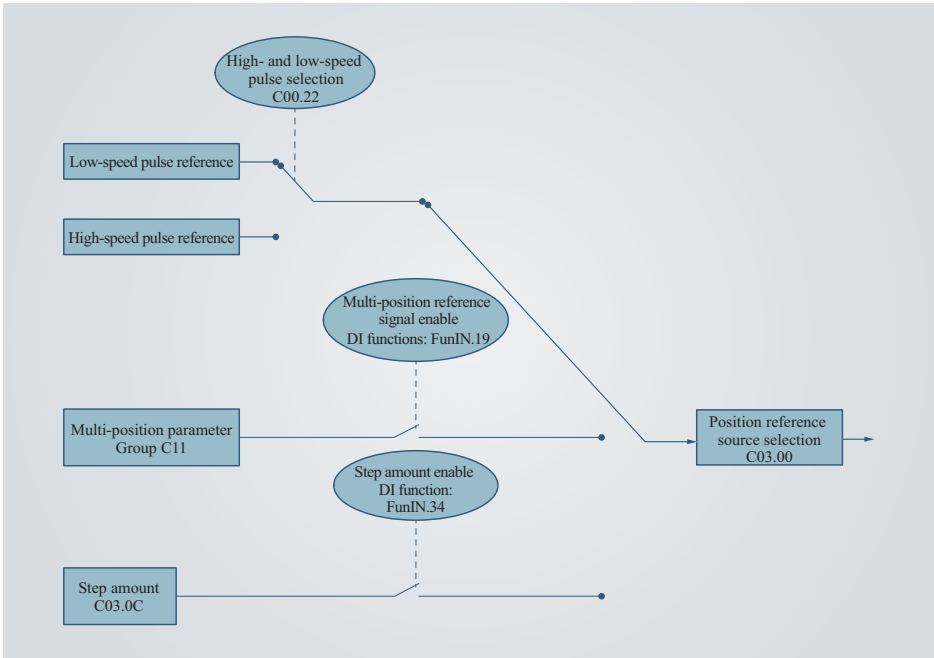


Figure 4-4 Position reference source setting

The position reference is a pulse reference (C03.00 = 0)

When selecting the pulse reference, perform the following steps to obtain the correct pulse reference form.

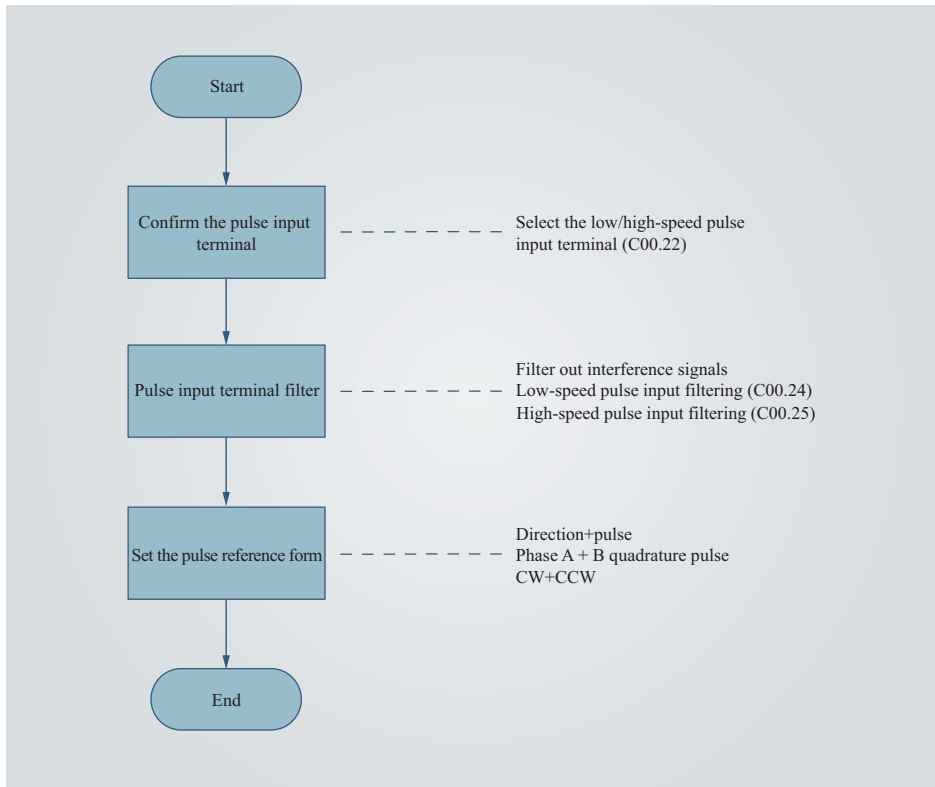
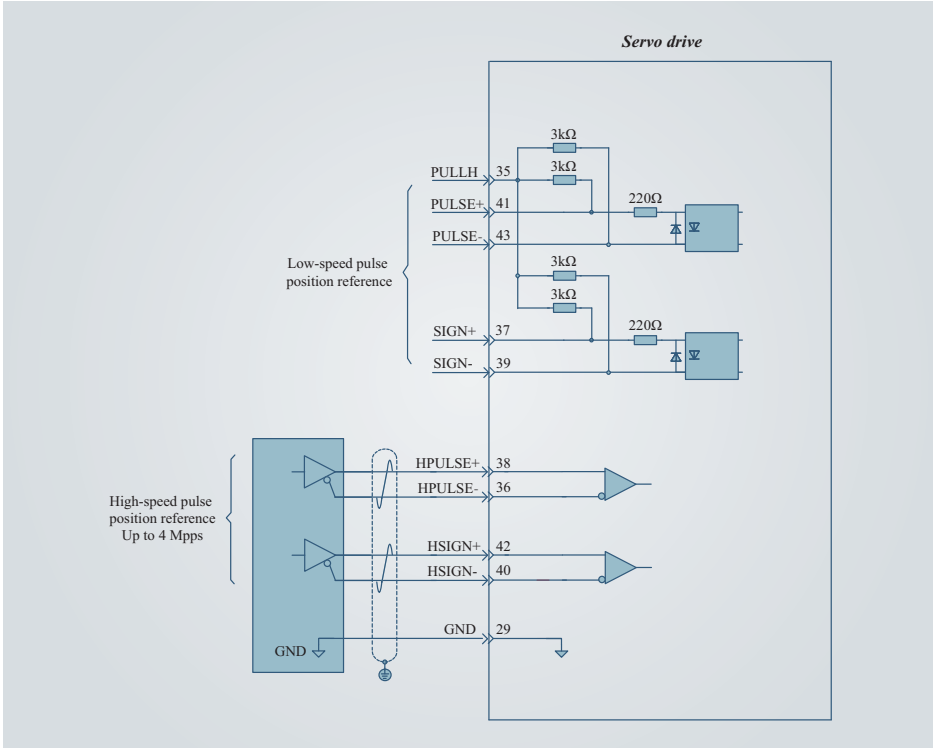


Figure 4-5 Pulse reference source setting flowchart

Pulse input terminal

The servo drive provides two pulse input terminals:



The low-speed pulse input terminals (PULSE+, PULSE-, SIGN+, and SIGN-) receive open-collector input (maximum frequency 200 kpps).

The high-speed pulse input terminals (HPULSE+, HPULSE-, HSIGN+, and HSIGN-) receive only differential input (maximum frequency 4 Mpps).

Table 4-1 Pulse input specification

Pulse		Maximum Frequency	Voltage	Forward Current
High-speed pulse	Differential signal	4 Mpps	5 V	<15 mA
Low-speed pulse	Differential signal	500 kpps	5 V	<15 mA
	Open-collector signal	200 kpps	24 V	<15 mA

Pulse input terminal filter

Set the filter time to filter the pulses from the low-speed/high-speed pulse input terminals to prevent motor malfunction due to interference on the servo drive.

If the filter time constant is t_f and the minimum width of input signals is t_{min} , the input signal and filtered signal are as shown in the following figure. The filtered signal has a t_f delay over the input signal.

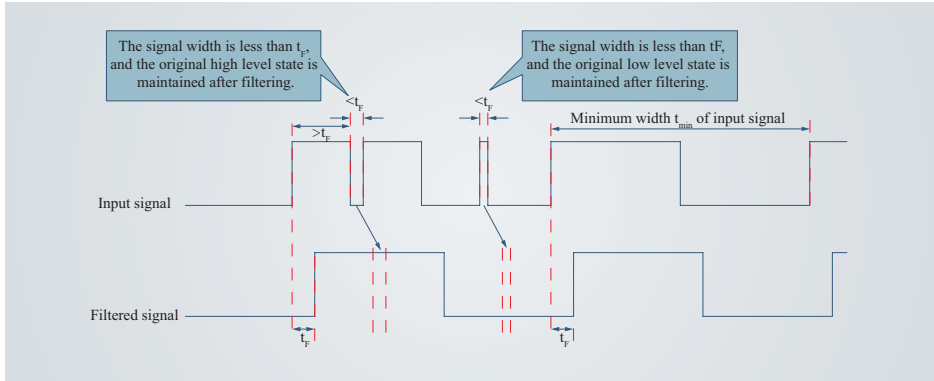


Figure 4-6 Schematic diagram of filtered signal waveform

The filter time constant t_f must meet the requirement: $t_f \leq (20\% - 25\%) t_{min}$.

The recommended filter parameter setting based on the maximum frequency (minimum width) of input pulses is described in the following table.

Table 4-2 Recommended Filter Time Constant

Pulse Input Terminal	Parameter	Maximum Frequency of Input Pulses	Recommended Filter Time Constant (Unit: ns)
Low-speed pulse input terminal	C00.24	< 167k	750
		167k to 200k	500
High-speed pulse input terminal	C00.25	200k to 1M	125
		> 1M	75

Pulse reference form

The servo drive supports three pulse input forms:

- Direction + pulse (positive or negative logic)
- Phase A + phase B quadrature pulse, 4-frequency multiplication
- CW+CCW (positive logic or negative logic)

Select the pulse input format appropriate for the host controller or other pulse output device.

C00.20	Pulse Form	Signal	Schematic Diagram of Forward Pulse	Schematic Diagram of Reverse Pulse
0	Pulse + direction Positive logic	PULSE SIGN		
1	Pulse + direction Negative logic	PULSE SIGN		
2	Phase A + B quadrature pulse 4-frequency multiplication	PULSE (Phase A) SIGN (Phase B)	Phase A leads phase B by 90°	Phase B leads phase A by 90°
3	CW + CCW Positive logic	PULSE (CW) SIGN (CCW)		
4	CW + CCW Negative logic	PULSE (CW) SIGN (CCW)		

The following table describes the maximum frequencies and widths of position pulses from different terminals.

Table 4-3 Pulse reference specification

Input Terminal		Maximum Frequency	Minimum Time Width (μs)			
			t ₁	t ₂	t ₃	t ₄
High-speed pulse input terminal		4 Mpps	0.125	0.125	0.125	0.25
Low-speed pulse input terminal	Differential	500 kpps	1	1	1	1
	Open collector	200 kpps	2.5	2.5	2.5	5

The rising time and falling time of position pulses must be smaller than 0.1 μs.

Multi-position reference as the position reference source (C03.00 = 1)

The servo drive supports multi-position running. It stores 16 position references, and the displacement, maximum running speed, and acceleration/deceleration time of each position reference can be set respectively. The waiting time and switch mode between positions can also be set according to actual requirements. The setting flowchart is as follows:

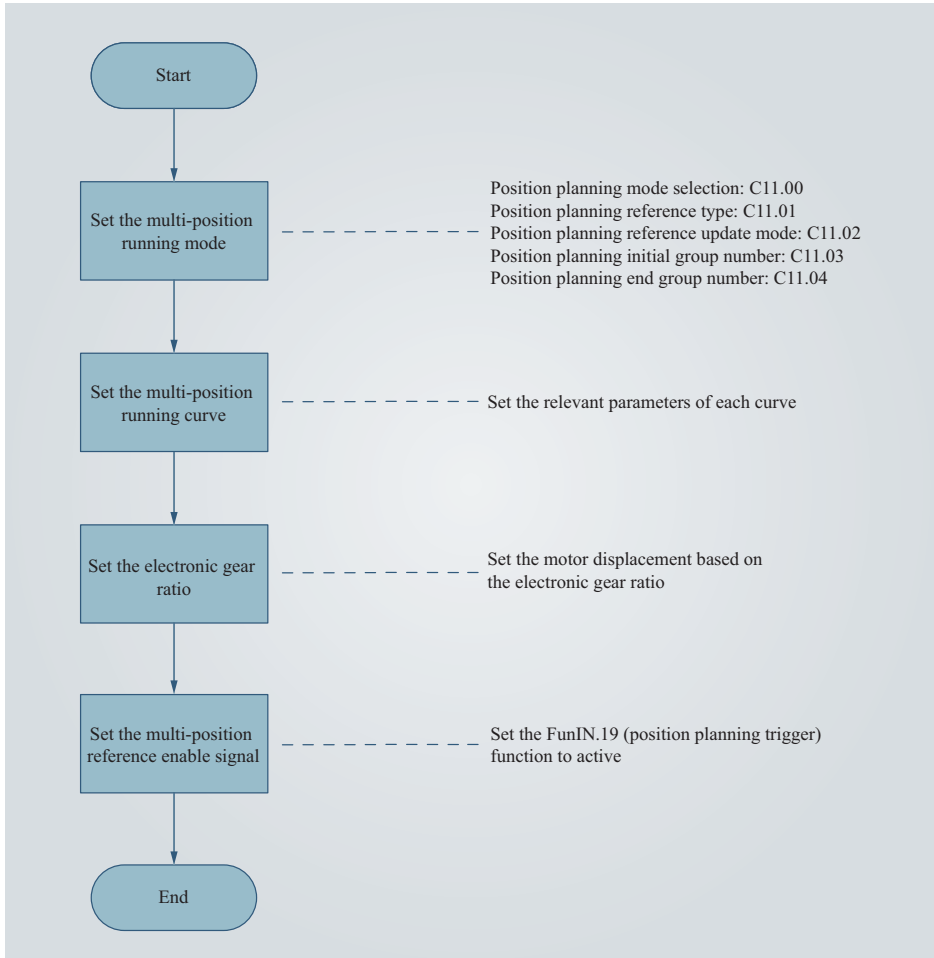
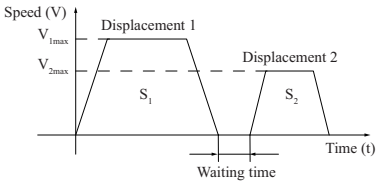


Figure 4-7 Multi-position reference source setting flowchart

Multi-position running mode

1. Individual running (C11.00 = 0)

Table 4-4 Individual running description

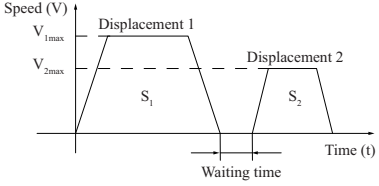
Mode Description	Running Curve
- The drive stops after one cycle of running. - The drive switches to the next displacement automatically. - The interval time between displacements can be set as needed. - The position planning trigger (PosInSen) signal is level triggered.	 Speed (V) Displacement 1 Displacement 2 Time (t) Waiting time V_{1max}, V_{2max}: maximum running speeds in displacement 1 and displacement 2 S_1, S_2: displacement 1 and displacement 2 - The positioning completed signal is active after each displacement is reached. - If the multi-position reference enable signal is switched off during operation, the drive abandons the unfinished displacement and stops. The COIN (positioning completed) signal is activated after the drive stops. - If the S-ON signal is switched off during operation, the motor stops according to the stop mode at S-ON OFF. The COIN (positioning completed) signal is deactivated after the motor stops. - Position planning can be paused through the DI function (FunIN.20: position planning pause).

NOTICE

- A cycle of running refers to the process that the drive runs from the initial position group number (C11.03) to the end position group number (C11.04).

2. Cyclic running (C11.00 = 1)

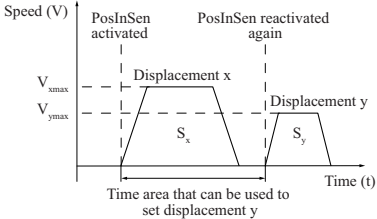
Table 4-5 Cyclic running description

Mode Description	Running Curve
• The drive starts from displacement 1 again after each cycle of running. • The drive switches to the next displacement automatically. • The interval time between displacements can be set as needed. • The cyclic running mode is kept when the FunIN.19 (position planning trigger) is active. • The position planning trigger (PosInSen) signal is level triggered.	 V_{1max}, V_{2max}: maximum running speeds in displacement 1 and displacement 2 S_1, S_2: displacement 1 and displacement 2 • The positioning completed signal is active after each displacement is reached. • If the multi-position reference enable signal is turned off during running, the servo drive discards the uncompleted displacement and stops. After the stop process is completed, the positioning completion signal is active. • If the S-ON signal is switched off during operation, the motor stops according to the stop mode at S-ON OFF. The COIN (positioning completed) signal is deactivated after the motor stops. • Position planning can be paused through the DI function (FunIN.20: position planning pause).

3. DI switchover running (C11.00 = 2)

When the multi-position running mode is DI switchover running, assign functions 12 to 15 (FunIN.12: CMD1 to FunIN.15: CMD4, multi-reference switchover) to four DIs of the servo drive, and set the active logic of these DIs.

Table 4-6 DI switchover running description

Mode Description	Running Curve
- The displacement to be executed next can be set when the current displacement is in progress. The motor stops after the current displacement is done executing. After the multi-position reference enable signal is switched on again, the present displacement will be executed. - The displacement No. is determined by the DI terminal logic. - There is no waiting time between displacements. The interval time between displacements is determined by the command delay of the host controller. - The position planning trigger (PosInSen) signal is edge triggered.	 V_{xmax}, V_{ymax}: maximum running speeds in displacement X and displacement Y S_x, S_y: displacement X and displacement Y - The positioning completed signal is active after each displacement is reached. - If the multi-position reference enable signal is switched off during operation, the drive continues to execute the unfinished displacement and outputs the COIN (positioning completed) signal. - The displacements must be switched in the following sequence: - Wait until displacement x is done executing before switching the displacement No. - When displacement x is in progress or done, switch off the multi-position reference enable signal first, and then change the displacement No. from x to y (if x = y, the drive executes displacement x again). - After displacement x is done executing, switch on the multi-position reference enable signal again to make the drive execute displacement y. - If the S-ON signal is switched off during operation, the motor stops according to the stop mode at S-ON OFF. The COIN (positioning completed) signal is deactivated after the motor stops. - Position planning can be paused through the DI function (FunIN.20: position planning pause).

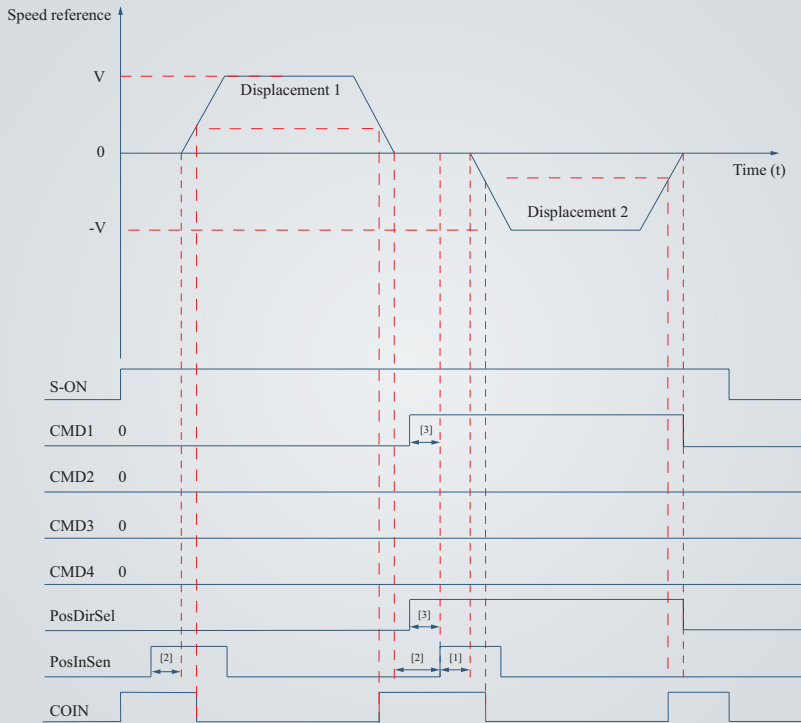


Figure 4-8 Multi-position sequence diagram

NOTICE

- The PosInSen signal is edge triggered. When using a normal DI terminal, ensure the effective signal width is at least 3 ms; when using a high-speed DI terminal, ensure the effective signal width is at least 0.25 ms.
- Area for switching the displacement No.: Refers to the range that starts from the moment the last position reference is done transmitting to the moment the next PosInSen signal is activated again.
- When a normal DI is used, an effective signal width of 0.125 ms must be kept.

Related functions:

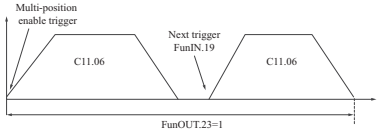
Code	Name	Function Name	Function
FunIN.12	CMD1	Multi-reference switchover 1	The position No. is a 4-bit binary value, and the relationship between the displacement No. and CMD1 to CMD4 is shown in the following table. The DI terminal logic is level triggered. The CMD value is 1 upon active level input or 0 upon inactive level input.
FunIN.13	CMD2	Multi-reference switchover 2	
FunIN.14	CMD3	Multi-reference switchover 3	
FunIN.15	CMD4	Multi-reference switchover 4	

Table 4-7 Relationship between the displacement No. and CMD1 to CMD4

CMD4	CMD3	CMD2	CMD1	Displacement No.
0	0	0	0	1
0	0	0	1	2
...				
1	1	1	1	16

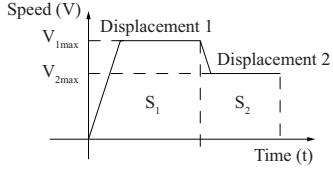
4. PP mode running (C11.00 = 3)

Table 4-8 Description of PP mode running

Mode Description	Running Curve
- The drive executes displacement 1 only. - The position planning trigger (PosInSen) signal is edge triggered.	 - Trigger a servo multi-position enable (FunIN.19). The servo drive stops after running for a distance defined by C11.06. - Position planning can be paused through the DI function (FunIN.20: position planning pause).

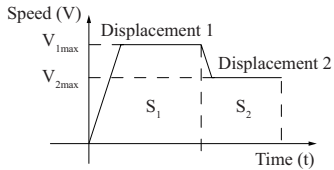
5. Single continuous running (C11.00 = 4)

Table 4-9 Description of single continuous running

Mode Description	Running Curve
• The drive stops after running for one cycle. • The drive switches to the next displacement automatically. • There is no waiting time between displacements.	 Speed (V) Displacement 1 Displacement 2 V_{1max} V_{2max} S_1 S_2 Time (t) V_{1max}, V_{2max}: maximum running speeds in displacement 1 and displacement 2 S_1, S_2: displacement 1 and displacement 2 • The positioning completed signal is active after each displacement is reached. • If the multi-position reference enable signal is switched off during operation, the drive abandons the unfinished displacement and stops. The COIN (positioning completed) signal is activated after the drive stops. • If the S-ON signal is switched off during operation, the motor stops according to the stop mode at S-ON OFF. The COIN (positioning completed) signal is deactivated after the motor stops. • Position planning can be paused through the DI function (FunIN.20: position planning pause).

6. Cyclic continuous running (C11.00 = 5)

Table 4-10 Description of cyclic continuous running

Mode Description	Running Curve
- The drive starts from displacement 1 again after each cycle of running. - The drive switches to the next displacement automatically. - There is no waiting time between displacements. - The position planning trigger (PosInSen) signal is edge triggered. - The position planning trigger (PosInSen) signal is active, and the drive maintains the cyclic running state.	 V_{1max}, V_{2max}: maximum running speeds in displacement 1 and displacement 2 S_1, S_2: displacement 1 and displacement 2 - The positioning completed signal is active after each displacement is reached. - If the multi-position reference enable signal is switched off during operation, the drive abandons the unfinished displacement and stops. The COIN (positioning completed) signal is activated after the drive stops. - If the S-ON signal is switched off during operation, the motor stops according to the stop mode at S-ON OFF. The COIN (positioning completed) signal is deactivated after the motor stops. - Position planning can be paused through the DI function (FunIN.20: position planning pause).

Multi-position running curve

A total of 16 position references can be set during multi-position operation. The displacement, maximum running speed, acceleration/deceleration time, and interval time between displacements can be set separately. The following table takes displacement 1 as an example.

The actual motor running curve according to the preceding setting is shown in the following figure.

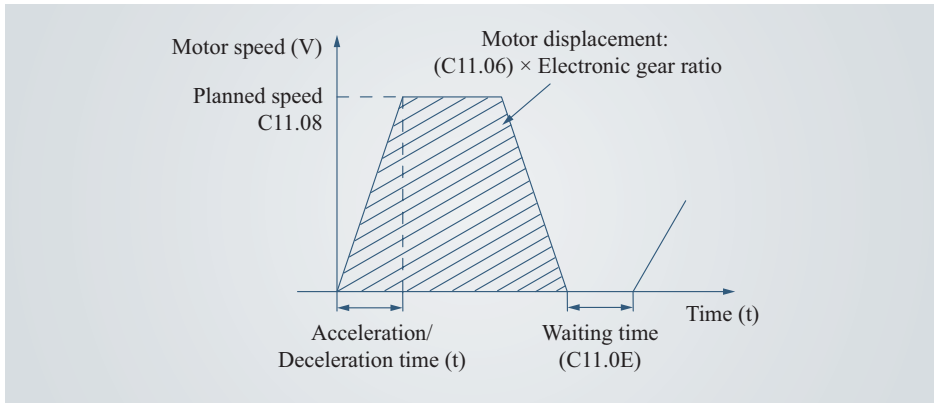


Figure 4-9 Motor running curve in displacement 1

Position planning trigger and pause

When using multi-position reference as the position reference source, assign function 19 (FunIN19: position planning trigger) and function 20 (FunIN20: position planning pause) to two DIs of the servo drive, and set the active logic of these DIs.

Code	Function Name	Function
FunIN.19	Position planning trigger	Active: The servo motor executes the multi-position planning. Inactive: The servo motor is in locked state.
FunIN.20	Position planning disable	Active: The servo motor pauses execution of the multi-position planning. Inactive: The servo motor executes the multi-position planning.

Step amount as the position reference source (C03.00 = 2)

NOTICE

- When the servo drive is in running state (S-ON signal active), the motor is in locked state if the step amount reference enable signal is inactive; the server motor runs if the step amount reference enable signal is active. When the execution of the C03.0C reference is completed and the step amount reference enable signal is no longer triggered, the motor is in locked state.

The servo drive also has the step amount running function, which means that the servo drive runs according to the internal fixed speed until the set displacement is reached. The setting flowchart is as follows:

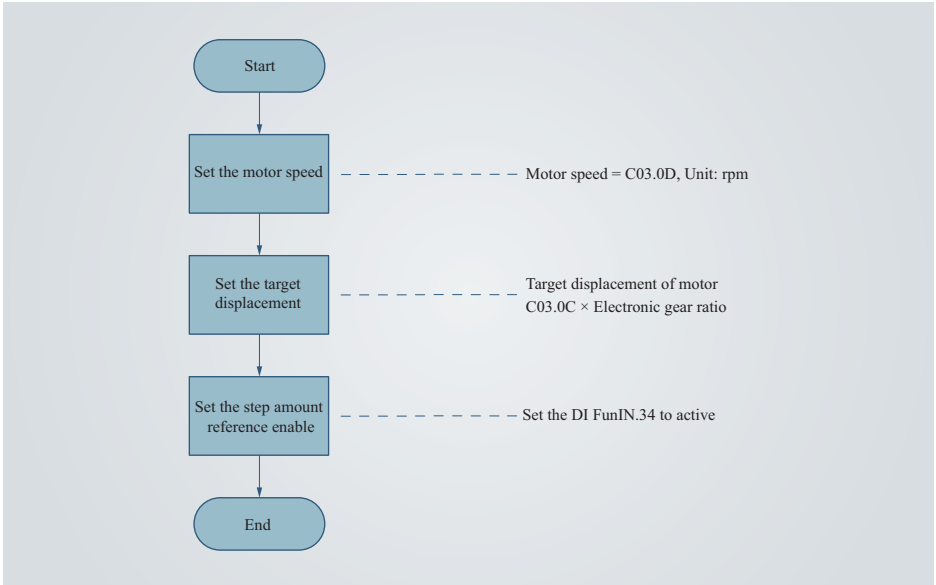


Figure 4-10 Flowchart for setting step amount reference as the position reference source

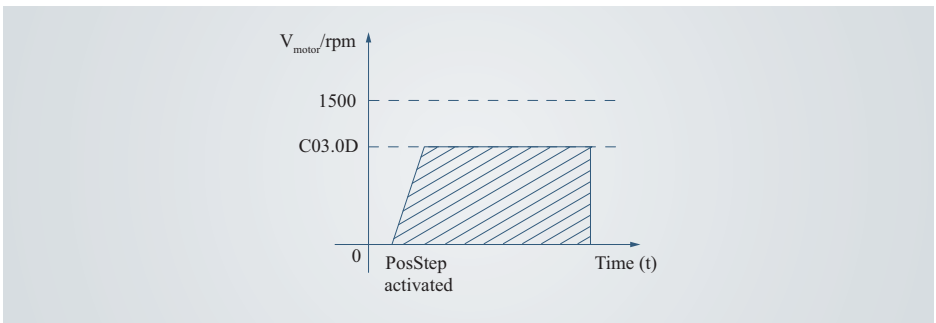


Figure 4-11 Motor running curve when C03.00 = 2

The hatched area in the preceding figure indicates the motor displacement: C03.0C x Electronic gear ratio, and the motor speed cannot exceed 1500 rpm.

- Motor displacement

When the position reference source is the step amount, the position reference sum (reference unit) is set in C03.0C. The sign of the C03.0C value determines whether the motor speed is a positive or negative value.

Parameter	Name	Value Range	Default	Unit	Modification Mode
C03.0C	Step amount reference value	-9999 to 9999	50	Unit in application	During operation

- Step amount reference enable

When using the step amount as the position reference source, assign function 34 (FunIN.34: PosStep, step amount reference enable) to a DI of the servo drive, and set the active logic of the DI.

Related functions:

Code	Name	Function Name	Function
FunIN.34	PosStep	Step amount reference enable	Servo running state Active: The position reference defined by C03.0C is input to the servo drive, driving the motor to run. Inactive: The servo motor is in locked state.

FunIN.34 (Step amount reference enable) is edge-triggered. The motor is locked after the step amount reference is done executing. When FunIN.34 is triggered again, the motor executes the step amount reference defined by C03.0C again.

Position reference direction

A DI can be used to change the position reference direction, so as to change the motor direction of rotation. Assign function 16 (FunIN.16: position reference direction) to a DI of the servo drive, and set the active logic of this DI.

Code	Function Name	Function
FunIN.16	Position reference direction	Inactive: Actual position reference direction same as the set direction. Active: Actual position reference direction opposite to the set direction.

NOTICE

- The actual motor direction is related to the rotating direction in C00.01, positive/negative of position reference, position reference direction (FunIN.16).

Table 4-11 Motor direction of rotation

C00.01	Sign of Position Reference	FunIN.16	Actual Motor Direction of Rotation
0	+	Inactive	CCW
0	+	Active	CW
0	-	Inactive	CW
0	-	Active	CCW
1	+	Inactive	CW
1	+	Active	CCW
1	-	Inactive	CCW
1	-	Active	CW

Position reference inhibited

The servo drive provides the position reference inhibited function (FunIN.24).

Position reference inhibited

The servo drive sets all the position references to 0, which means it does not respond to any internal or external position references, and the motor is in the locked state in the position control mode. In this case, the drive can switch to other control modes to continue operating. To use this function, assign function 24 (FunIN.24: position reference inhibited) to a DI of the servo drive, and set the active logic of this DI. The high-speed DI terminal (DI7 or DI8) is recommended.

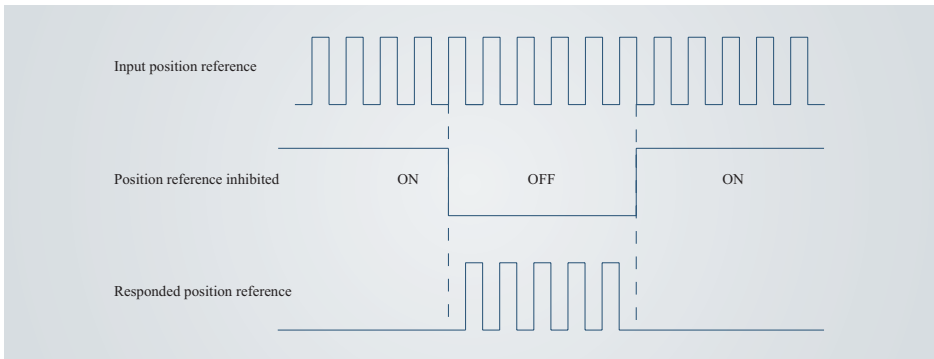


Figure 4-12 Waveform example for position reference inhibited

Related functions:

Code	Function Name	Function
FunIN.24	Position reference inhibited	Inactive: The drive responds to position references in position control mode. Active: The drive does not respond to any internal or external position references in position control mode.

NOTICE

- When DI is used, keep an interval of at least 0.5 ms from the moment the DI logic is deactivated to the moment the internal position reference is inputted.

4.1.3 Reference Frequency Division/Multiplication (Electronic Gear Ratio)***Definition of the electronic gear ratio***

In the position control mode, the input position reference (reference unit) defines the load displacement; the motor position reference (encoder unit) defines the motor displacement. The electronic gear ratio is used to establish a proportional relationship between the input position reference and motor position reference.

The electronic gear ratio, which allows frequency division (electronic gear ratio < 1) or frequency multiplication (electronic gear ratio > 1), can be used to set the actual displacement corresponding to the input position reference per reference unit, or used to increase the position reference frequency when the motor speed needed cannot be fulfilled due to limited pulse output frequency of the host controller or limited parameter value range.

NOTICE

- Reference unit: Refers to the minimum identifiable value input from the host controller to the servo drive.
- Encoder unit: Refers to the value of the input reference multiplied/divided by the electronic gear ratio.

Procedure for setting the electronic gear ratio

The electronic gear ratio varies according to the mechanical structure. Set it according to the following flowchart.

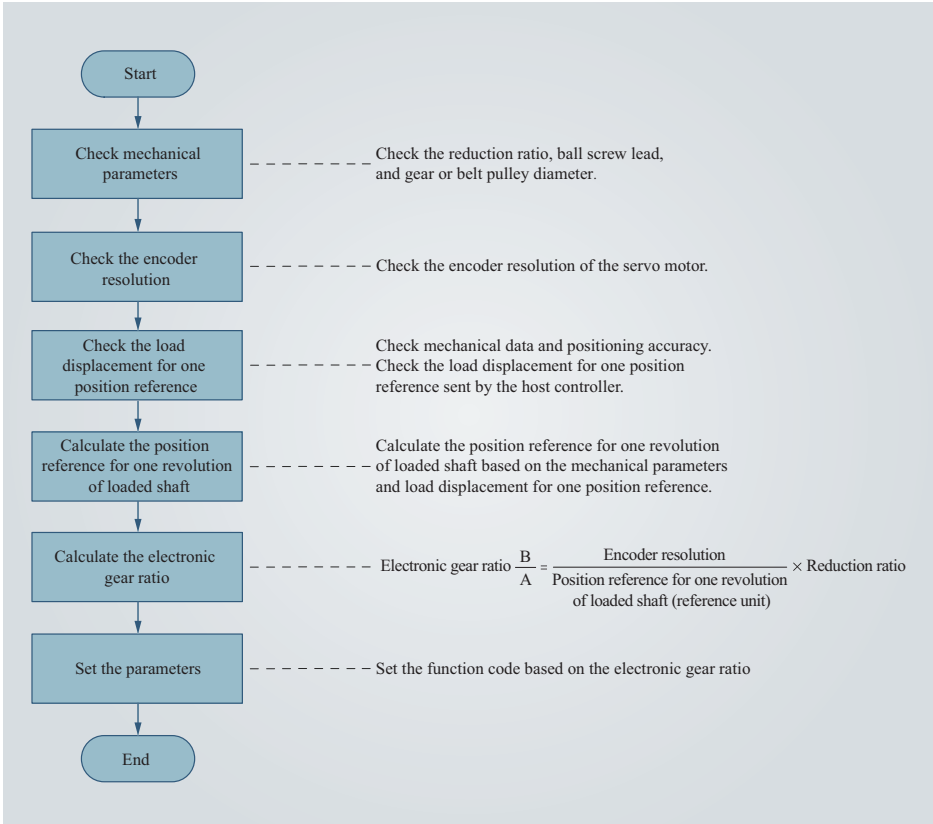


Figure 4-13 Procedure for setting the electronic gear ratio

See the following figure for how to set parameters.

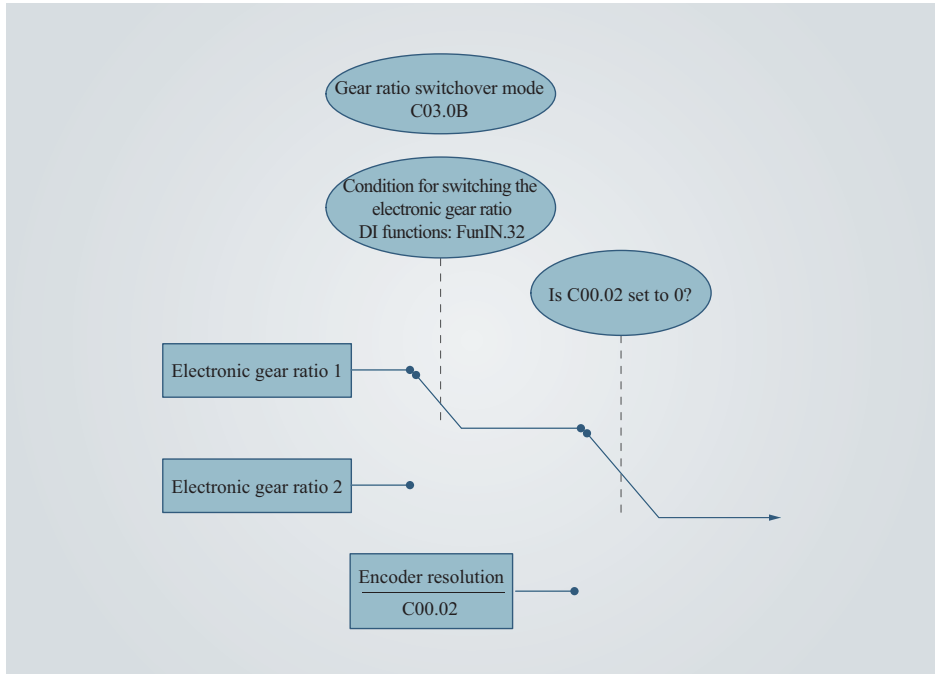


Figure 4-14 Flowchart for setting the electronic gear ratio

NOTICE

- When C00.02 is not equal to 0, the following formula applies

$$\text{Electronic gear ratio } \frac{B}{A} = \frac{\text{Encoder resolution}}{C00.02}$$

In this case, electronic gear ratios 1 and 2 are invalid.

⚠ CAUTION

- The motor speed may fluctuate significantly if the electronic gear ratio changes sharply in real time or electronic gear ratio 1 differs greatly from electronic gear ratio 2. In this case, use the position reference filter function (C01.20, C01.21, C01.22, or C01.23) to allow smooth switchover of position references.

Switching the electronic gear ratio

- The electronic gear ratio can be switched when C00.02 is set to 0. Determine whether it is necessary to switch between electronic gear ratios 1 and 2 based on mechanical conditions. Set the condition for switching the electronic gear ratio.
- Only one electronic gear ratio is effective at a moment.
- The effective time of real-time change in the electronic gear ratio is also restricted by the switchover condition.

Assign function 32 (FunIN.32: electronic gear ratio switchover) to a DI of the servo drive, and set the active logic of this DI.

Related functions:

Code	Function Name	Function
FunIN.32	Condition for switching the electronic gear ratio	Inactive: Electronic gear ratio 1 is selected in position control mode. Active: Electronic gear ratio 2 is selected in position control mode.

The electronic gear ratio selected by the servo drive is described in the following table.

C00.02	C03.0B	Level of DI Assigned with FunIN.32	Electronic gear ratio
0	0	Inactive	C03.02/C03.04
		Active	C03.06/C03.08
	1	Inactive	C03.02/C03.04
		Active	C03.06/C03.08
1 to 1048576	-		

The resolution of the serial encoder is 2^n (P/r), where "n" is the number of bits of the serial encoder.

Calculating the electronic gear ratio

The following figure shows the relationship among the position reference (reference unit), load displacement, and electronic gear ratio:

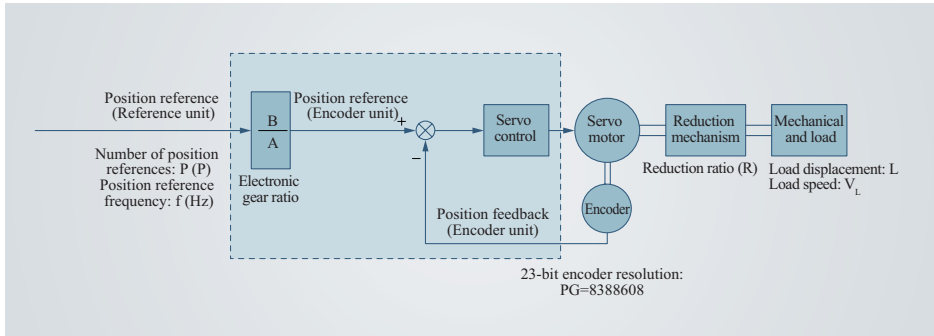


Figure 4-15 Relationship among the position reference (reference unit), load displacement, and electronic gear ratio

Take the ball screw in linear motion as an example, with p_B (mm) as the screw lead, P_G as the encoder resolution, and R as the reduction ratio of the reduction mechanism.

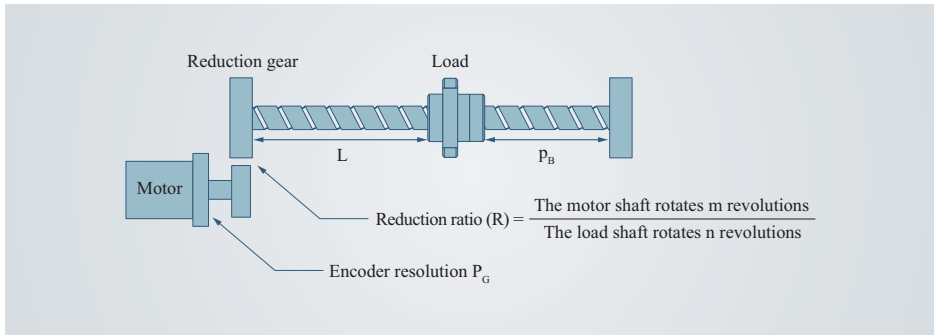


Figure 4-16 Ball screw

- When the load displacement per pulse ΔL (mm) is known:

When the load displacement is ΔL , the load shaft rotates $\frac{\Delta L}{p_B}$ revolutions, and the motor shaft rotates $\frac{\Delta L}{p_B} \times R$ revolutions.

$$1 \times \frac{B}{A} = \frac{\Delta L}{p_B} \times R \times P_G$$

Therefore, the electronic gear ratio is calculated as follows:

$$\frac{B}{A} = \frac{\Delta L}{p_B} \times R \times P_G$$

- When the load displacement L (mm) and position reference sum P (P) are known:

When the mechanical displacement is L , correspondingly the load shaft rotates $\frac{L}{p_B}$ revolutions and the motor shaft rotates $\frac{L}{p_B} \times R$ revolutions.

$$P \times \frac{B}{A} = \frac{L}{p_B} \times R \times P_G$$

Therefore, the electronic gear ratio is calculated as follows:

$$\frac{B}{A} = \frac{L}{p_B} \times R \times P_G \times \frac{1}{P}$$

- When the load moving speed V_L (mm/s) and position reference frequency f (Hz) are known:

Load shaft speed: $\frac{V_L}{p_B}$ (r/s)

Motor speed: $V_M = \frac{V_L}{p_B} \times R$

The relationship among the position reference frequency, electronic gear ratio, and motor speed is as follows:

$$f \times \frac{B}{A} = V_M \times P_G$$

Therefore, the electronic gear ratio is calculated as follows:

$$\frac{B}{A} = \frac{V_M \times P_G}{f}$$

Example for setting the electronic gear ratio

Step	Name	Mechanical Structure		
		Ball Screw Transmission	Belt Pulley Transmission	Rotating Load
1	Mechanical parameters	Reduction ratio (R): 1/1 Screw lead: 0.01m	Reduction ratio (R): 5/1 Diameter of belt pulley: 0.2m (Circumference of belt pulley: 0.628m)	Reduction ratio (R): 10/1 Load angle of rotation per revolution of the load shaft: 360°
2	Encoder resolution	17bit=131072P/r	17bit=131072P/r	17bit=131072P/r
3	Load displacement per position reference (reference unit)	0.0001m	0.000005m	0.01°
4	Position references per revolution of the load shaft (reference unit)	$\frac{0.01}{0.0001} = 100$	$\frac{0.628}{0.000005} = 125600$	$\frac{360}{0.01} = 36000$
5	Calculation	$\frac{B}{A} = \frac{131072}{100} \times \frac{1}{1}$	$\frac{B}{A} = \frac{131072}{125600} \times \frac{5}{1}$	$\frac{B}{A} = \frac{131072}{36000} \times \frac{10}{1}$

Step	Name	Mechanical Structure		
		Ball Screw Transmission	Belt Pulley Transmission	Rotating Load
6	Setting	C03.02=131072 C03.04=100	C03.02=655360 C03.04=125600	C03.02=1310720 C03.04=36000

4.1.4 Position reference filter

This function filters the position references (encoder unit) divided or multiplied by the electronic gear ratio. Position reference filter includes first-order low-pass filter and moving average filter.

It is applicable to the following conditions:

- The acceleration/deceleration process is not performed on the position references sent from the host controller.
- The pulse frequency is too low.
- The electronic gear ratio is larger than 10.

This function has no effect on the displacement (position reference sum).

If the setting is excessive, the response delay may be too large. Therefore, set the filter time constant based on actual conditions.

4.1.5 Position deviation clear

Position deviation = Position reference sum - Position feedback sum

The drive clears the position deviation when the position deviation clear function is enabled (C03.16=1) and DI (FunIN.23: position deviation clear) is active.

To use this function, assign function 23 (FunIN.23: position deviation clear) to a DI of the servo drive, and set the active logic of this DI. The high-speed DI terminal (DI7 or DI8) is recommended.

Related functions:

Code	Function Name	Function
FunIN.23	Position deviation clear	Active: Clear the position deviation. Inactive: Do not clear the position deviation.

4.1.6 Frequency-division output

The frequency-division output function outputs the position reference pulses or encoder feedback position references as A/B phase quadrature pulses.

Code	Function Name	Function
C00.2A	Z-pulse output polarity selection	0: Output differential Z-pulse with positive polarity, and OCZ pulse with positive polarity. 1: Output differential Z-pulse with positive polarity, and OCZ pulse with negative polarity. 2: Output differential Z-pulse with negative polarity, and OCZ pulse with positive polarity. 3: Output differential Z-pulse with negative polarity, and OCZ pulse with negative polarity.

NOTICE

- It is recommended to use the active edge output by the Z signal when a high precision frequency-division output of Z signal is required.

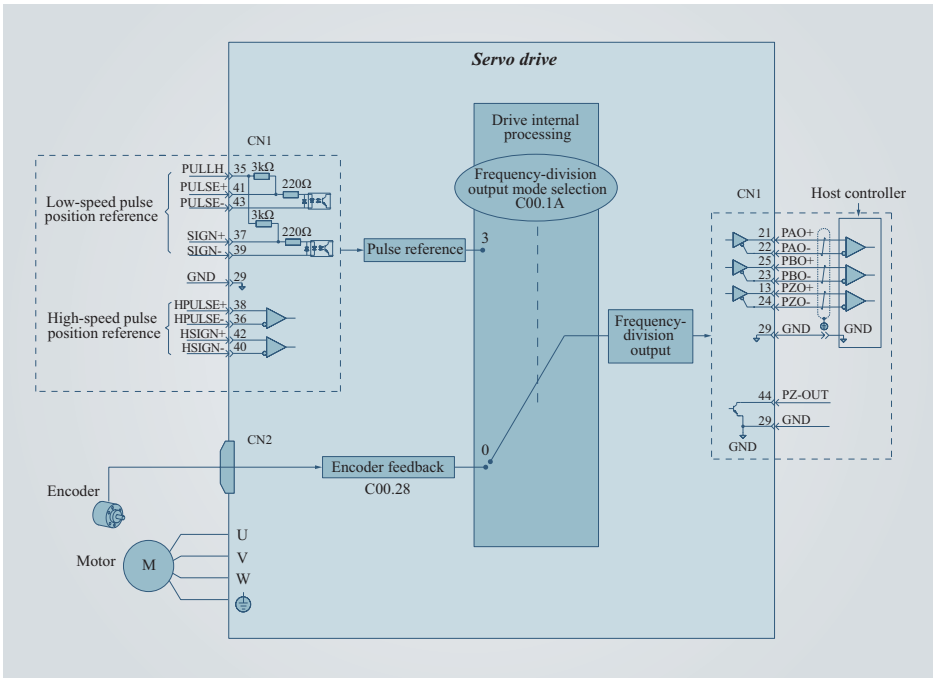


Figure 4-17 Schematic diagram of frequency-division output

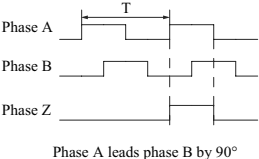
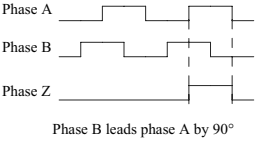
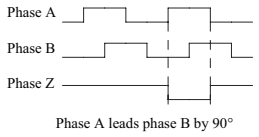
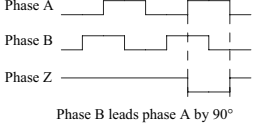
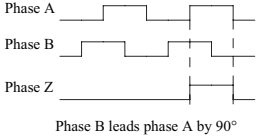
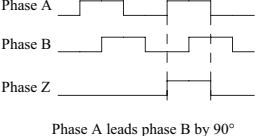
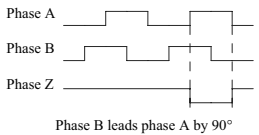
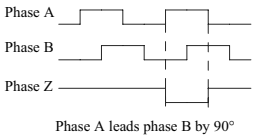
It is recommended to use synchronous output (C00.1A=3) of pulse references in case of synchronous tracing of multi-axis servo pulses. When the host controller is used for closed-loop feedback, it is recommended to use encoder frequency-division output (C00.1A=0). The servo drive provides one group of frequency-division terminals, as described below:

- Phase A pulse: PAO+, PAO-, differential output, maximum output pulse frequency 4 Mpps
- Phase B pulse: PBO+, PBO-, differential output, maximum output pulse frequency 4 Mpps
- Phase Z pulse: PZO+, PZO-, differential output, maximum output pulse frequency 4 Mpps
- PZ OUT, GND, open-collector output, maximum output pulse frequency: 200 kpps

When using the frequency-division output function, set the output pulse source (C00.1A), phase (C00.27), resolution (C00.28), and phase Z pulse polarity (C00.2A) according to requirements.

When the output source is encoder frequency-division pulse (C00.1A=0), the phase A/B output pulses per motor revolution are determined by C00.28. The pulse width (T) of phase A/B is determined by the motor speed. The phase Z, whose width is also T, is synchronized with phase A. The Z signal is output once per motor revolution.

Table 4-12 Encoder frequency-division output pulses (C00.1A = 0)

C00.27 (Output pulse phase)	C00.2A (Z pulse output polarity)	Pulse Output Diagram of Forward RUN	Pulse Output Diagram of Reverse RUN
0	0	 Phase A leads phase B by 90°	 Phase B leads phase A by 90°
	1	 Phase A leads phase B by 90°	 Phase B leads phase A by 90°
1	0	 Phase B leads phase A by 90°	 Phase A leads phase B by 90°
	1	 Phase B leads phase A by 90°	 Phase A leads phase B by 90°

4.1.7 Positioning completed/near function

Positioning completed: When the position deviation fulfills the condition set by users (C03.10), it indicates the positioning in position control mode is completed. Meanwhile, the servo drive outputs positioning

completed (COIN) signal, and the host controller, after receiving this signal, confirms that the positioning is completed.

The following figure shows the schematic diagram.

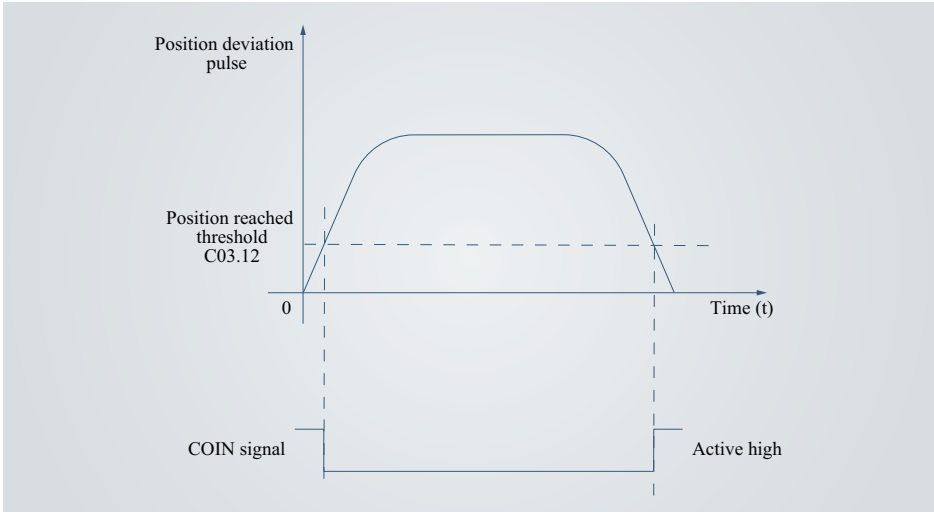


Figure 4-18 Description of the positioning completed/near function

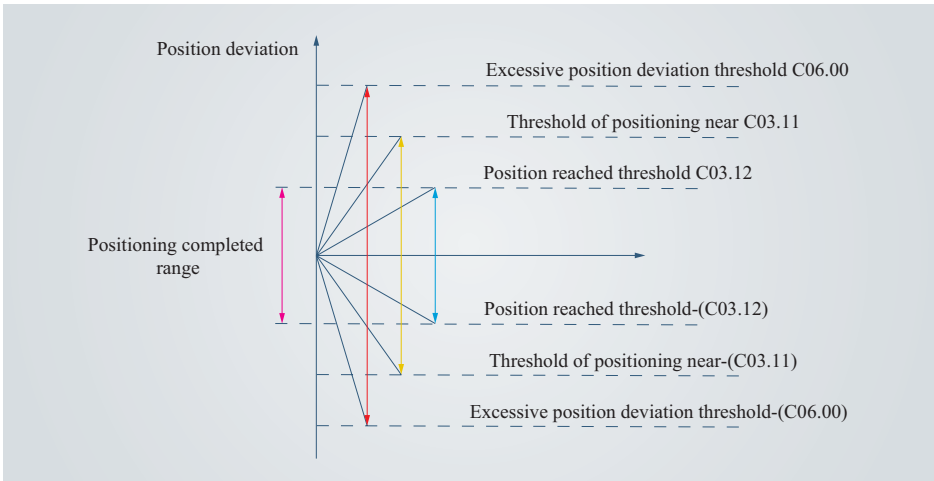


Figure 4-19 Position deviation-related signals

When the position deviation meets the condition defined by C03.10, the servo drive outputs the positioning near (NEAR) signal. The host controller sends the positioning near signal to the servo drive before confirming positioning completed, so that the servo drive prepares for the positioning completed action.

Before using the positioning completed/near function, set the output condition, threshold, window, and hold time. The schematic diagram for the window time and hold time of the positioning completed signal is as follows.

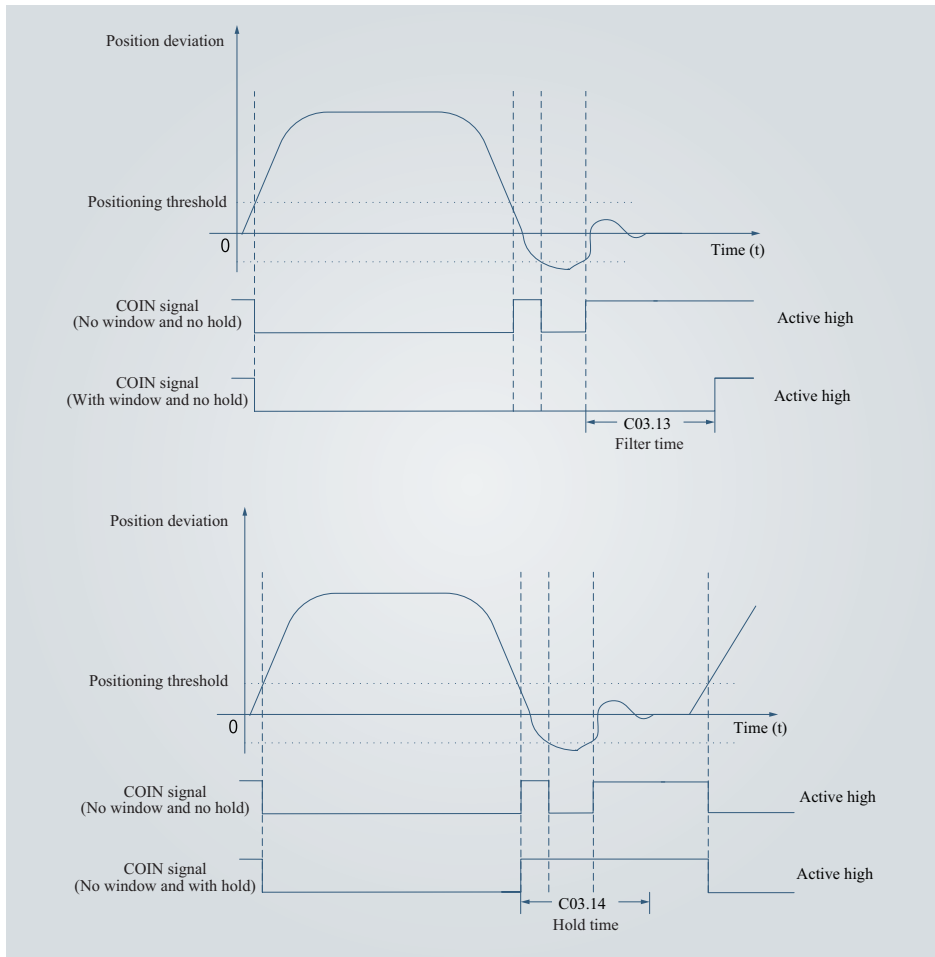


Figure 4-20 Schematic diagram for the window time and hold time of the positioning completed signal

When the COIN (positioning completed) signal has a hold time of 0, this signal remains active until the next position reference is received.

 CAUTION

- Set the position near threshold (C03.11) to a value higher than the position complete threshold (C03.12) in general cases.
- The position complete threshold (C03.12) reflects only the absolute threshold when the positioning completed signal is active. It is irrelevant to the positioning precision.
- An excessively high speed feedforward gain (C01.14) or low-speed operation reduces the absolute position deviation. In this case, the COIN (positioning completed) signal may keep active if C03.12 is set to an excessively high value. To improve the positioning accuracy, decrease the value of C03.12.
- When the position complete threshold (C03.12) is set to a low value along with small position deviation, you can set C03.11 to change the condition for outputting the COIN (positioning completed) signal.
- When the S ON signal is inactive, the positioning completed signal (COIN) and positioning near signal (NEAR) are inactive.
- To use the positioning completed/near function, assign function 7 (FunOUT.7: positioning completed) and function 8 (FunOUT.8: positioning near) to two DOs of the servo drive, and set the active logic of these DOs.

Related functions:

Code	Name	Function Name	Function
FunOut.7	COIN	Positioning completed	Active: The absolute position deviation meets the threshold defined by C03.12 in position control mode, indicating positioning is completed. Inactive: The servo drive is in the positioning completion process in position control mode.
FunOut.8	NEAR	Positioning near	Active: The absolute position deviation meets the condition defined by C03.11, indicating the servo drive is close to the target position. Inactive: The servo drive is in the positioning proximity process in position control mode.

4.1.8 Interrupt positioning

CAUTION

- The interrupt positioning signal cannot be triggered during homing.

Function description

If interrupt positioning is triggered in the position control mode, the servo drive halts current operation and turns to executing the pre-set fixed distance. To be specific, when the S-ON signal is active in position control mode, if this function is enabled, the servo motor runs the position reference for interrupt positioning in the original direction (before the function is triggered).

When interrupt positioning is in progress, the servo drive does not respond to any other internal/external position references (including another interrupt positioning command). After the running of this function is complete, the servo drive keeps shielding or responds to position references according to the setting of C10.21, but discards the position references input in the running process.

After interrupt positioning is done, the servo drive outputs the interrupt positioning completed (FunOUT.10) signal and positioning completed (FunOUT.7) signal, while the host controller, upon receiving the interrupt positioning completed signal, acknowledges that interrupt positioning is done. Output of the interrupt positioning completed signal is not related to the S-ON signal or the logic of DI7.

Interrupt positioning is active only when the following conditions are met:

- The motor speed is higher than or equal to 10 rpm before interrupt positioning is triggered, or C10.22 and C10.24 are not 0.
- The DI assigned with FunIN.21 (Interrupt positioning inhibited) is not used or the logic of this DI is inactive.

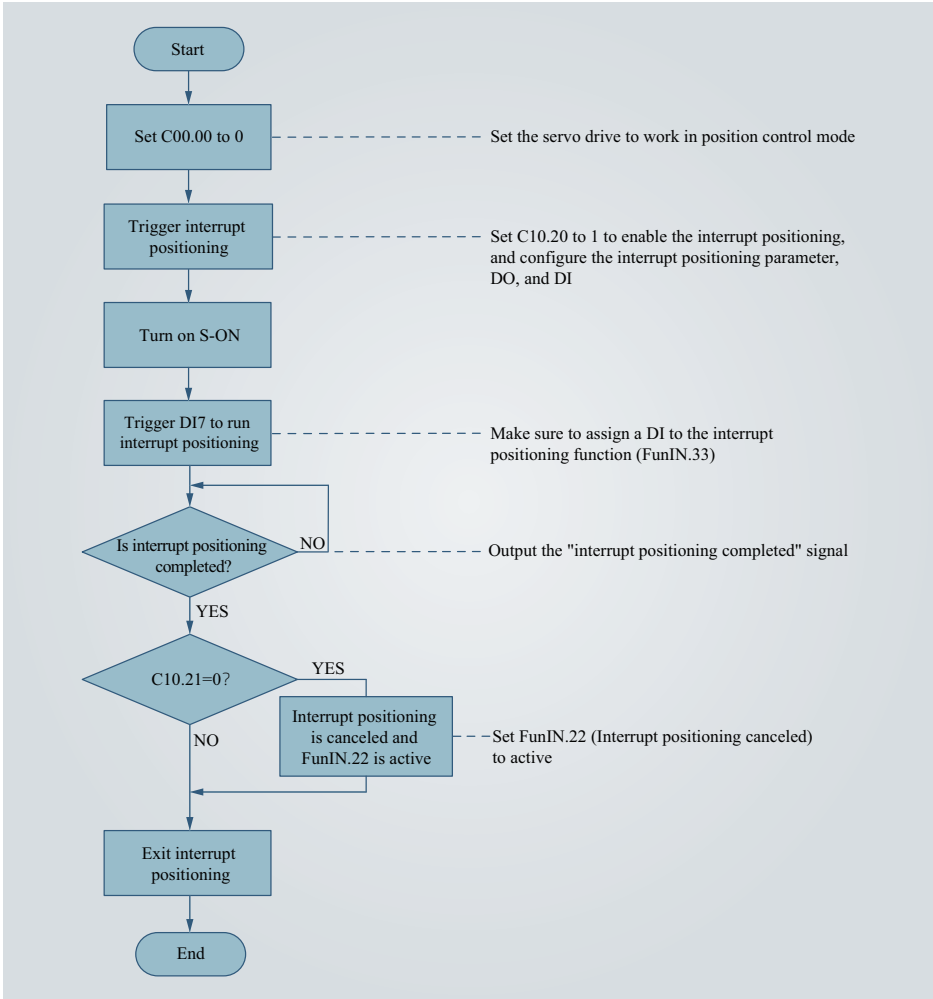
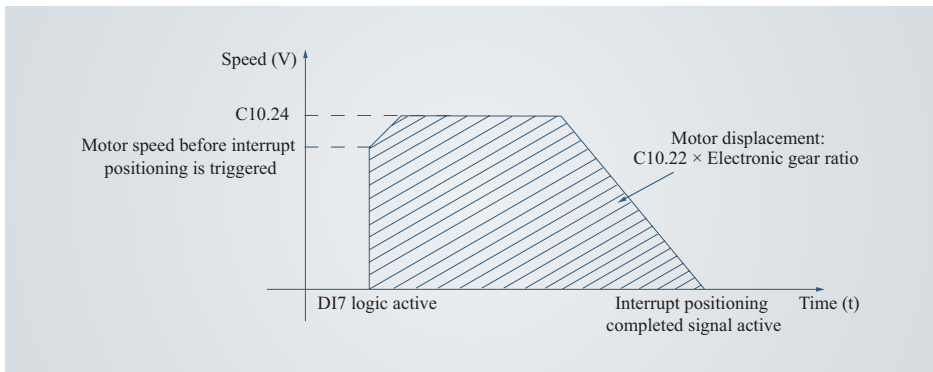


Figure 4-21 Flowchart of the interrupt positioning signal

Related functions:

Code	Function Name	Function
FunIN.22	Interrupt positioning canceled	Active: The interrupt positioning state is cleared, which means the servo drive can respond to other position references. Inactive: The interrupt positioning state is locked, which means the servo drive cannot respond to other position references.
FunIN.21	Interrupt positioning inhibited	Active: Interrupt positioning inhibited Inactive: Interrupt positioning allowed
FunIN.33	Interrupt positioning enable	Active: Interrupt positioning enabled Inactive: Interrupt positioning disable DI45 is assigned with the interrupt positioning enable function.
FunIN.10	Interrupt positioning completed	Active: Interrupt positioning is completed in position control mode. Inactive: Displacement in interrupt positioning is not completed in position control mode.

**Figure 4-22 Motor operating curve during interrupt positioning****Table 4-13 Motor speed during interrupt positioning**

C10.24	Motor Speed Before Interrupt Positioning Is Triggered (rpm)	Interrupt Positioning	Constant Operating Speed in Interrupt Positioning
0	<10	Inactive	-
	≥10	Active	Motor speed before interrupt positioning is triggered
1 to 6000	-	Active	C10.24

4.1.9 Homing

CAUTION

- The homing signal is shielded when the interrupt positioning or multi-position planning function is in progress.

Function description

- Home (or mechanical home): Indicates the position of the home switch or Z signal, which is determined by C10.01.
- Zero: positioning target point, represented as home + offset (set in C10.0B). When C10.0B is set to 0, the zero position coincides with the home.

In position control mode, when homing is triggered after the S-ON signal is activated, the servo motor starts searching for the zero position.

When homing is in progress, the servo drive does not respond to other position references (including another homing enable signal triggered) until homing is done. When homing is completed, the servo drive can respond to other position references.

- Home attaining: After receiving the homing signal, the servo drive proactively locates the relative position between the motor shaft and the preset mechanical home reference point; it finds the home and then moves through the offset from the home reference point to the zero point. The homing mode usually applies in initial searching for the zero position.

After the homing operation is completed, the absolute motor position (U40.16) is the same as the mechanical home offset (C10.0B).

After the homing operation is completed, the servo drive outputs the home attaining completed signal (FunOUT.9). After receiving this signal, the host controller confirms that the homing operation is completed. The home attaining completed signal is not related to the mode or running state of the servo drive.

Home attaining

CAUTION

- Set mechanical limit switches before enabling the homing function. For homing upon hit-and-stop, set the offset to a value within the travel range to prevent the machine from collision due to high-speed operation during homing.

Related parameters:

Parameter	Name	Value Range	Default	Unit	Function Description
C10.00	Homing enable	0 to 3	0	-	Set the homing enable mode.
C10.01	Homing method selection	-2 to 35	0	-	Set the homing mode.
C10.02	Initial homing speed	0 to 8000	100	rpm	Set the homing start speed.
C10.03	End homing speed	0 to 3000	10	rpm	Set the homing end speed.
C10.04	Homing acceleration time	0 to 3600000	1000	ms	Set the homing acceleration time.
C10.06	Homing deceleration time	0 to 3600000	1000	ms	Set the homing deceleration time.
C10.08	Homing timeout interval	0 to ($2^{32}-1$)	60000	ms	Set the homing timeout.
C10.0A	Homing offset mode	0 to 1	0	-	Set the homing offset mode.
C10.0B	Homing offset distance	-2^{31} to ($2^{31}-1$)	0	Unit in application	Set the homing offset distance.

Homing mode introduction:**Table 4-14 Mode overview table**

Mode setting	Description
-2	The mode is used to search for the mechanical extreme position and Z pulse in the forward direction.
-1	The mode is used to search for the mechanical extreme position and Z pulse in the reverse direction.
0	-
1	The motor starts operation in the reverse direction, switches to low-speed operation when the negative limit (NL) status changes from OFF to ON during reverse operation, and then retreats to search for the nearest Z pulse position as the home.
2	The motor starts operation in the forward direction, switches to low-speed operation when the positive limit (PL) status changes from OFF to ON during forward operation, and then retreats to search for the nearest Z pulse position as the home.
3	The motor operates in the forward direction when HSW is inactive and reverse direction when HSW is active. The motor switches to low-speed operation when the HSW status changes from ON to OFF during reverse operation, and then continues the reverse operation to search for the nearest Z pulse position as the home.

Mode setting	Description
4	The motor operates in the forward direction when HSW is inactive and reverse direction when HSW is active. The motor switches to low-speed operation when the HSW status changes from OFF to ON during forward operation, and then continues the forward operation to search for the nearest Z pulse position as the home.
5	The motor operates in the reverse direction when HSW is inactive and forward direction when HSW is active. The motor switches to low-speed operation when the HSW status changes from ON to OFF during forward operation, and then continues the forward operation to search for the nearest Z pulse position as the home.
6	The motor operates in the reverse direction when HSW is inactive and forward direction when HSW is active. The motor switches to low-speed operation when the HSW status changes from OFF to ON during reverse operation, and then continues the reverse operation to search for the nearest Z pulse position as the home.
7	The motor operates in the forward direction when HSW is inactive and reverse direction when HSW is active. The motor switches to low-speed operation when the HSW status changes from ON to OFF during reverse operation, and then continues the reverse operation to search for the nearest Z pulse position as the home.
8	The motor operates in the forward direction when HSW is inactive and reverse direction when HSW is active. The motor switches to low-speed operation when the HSW status changes from OFF to ON during forward operation, and then continues the forward operation to search for the nearest Z pulse position as the home.
9	The motor operates in the forward direction no matter whether HSW is active or inactive. The motor switches to low-speed operation when the HSW status changes from OFF to ON during reverse operation, and then continues the reverse operation to search for the nearest Z pulse position as the home.
10	The motor operates in the forward direction no matter whether HSW is active or inactive. The motor switches to low-speed operation when the HSW status changes from ON to OFF during forward operation, and then continues the forward operation to search for the nearest Z pulse position as the home.
11	The motor operates in the reverse direction when HSW is inactive and forward direction when HSW is active. The motor switches to low-speed operation when the HSW status changes from ON to OFF during forward operation, and then continues the forward operation to search for the nearest Z pulse position as the home.
12	The motor operates in the reverse direction when HSW is inactive and forward direction when HSW is active. The motor switches to low-speed operation when the HSW status changes from OFF to ON during reverse operation, and then continues the reverse operation to search for the nearest Z pulse position as the home.
13	The motor operates in the reverse direction no matter whether HSW is active or inactive. The motor switches to low-speed operation when the HSW status changes from OFF to ON during forward operation, and then continues the forward operation to search for the nearest Z pulse position as the home.
14	The motor operates in the reverse direction no matter whether HSW is active or inactive. The motor switches to low-speed operation when the HSW status changes from ON to OFF during reverse operation, and then continues the reverse operation to search for the nearest Z pulse position as the home.

Mode setting	Description
15	Reserved
16	Reserved
17	Similar to mode 1. During reverse operation, the position where the NL status changes from OFF to ON is used as the home, without searching for the Z pulse.
18	Similar to mode 2. During forward operation, the position where the PL status changes from OFF to ON is used as the home, without searching for the Z pulse.
19	Similar to mode 3. During reverse operation, the position where the HSW status changes from ON to OFF is used as the home, without searching for the Z pulse.
20	Similar to mode 4. During forward operation, the position where the HSW status changes from OFF to ON is used as the home, without searching for the Z pulse.
21	Similar to mode 5. During forward operation, the position where the HSW status changes from ON to OFF is used as the home, without searching for the Z pulse.
22	Similar to mode 6. During reverse operation, the position where the HSW status changes from OFF to ON is used as the home, without searching for the Z pulse.
23	Similar to mode 7. During reverse operation, the position where the HSW status changes from ON to OFF is used as the home, without searching for the Z pulse.
24	Similar to mode 8. During forward operation, the position where the HSW status changes from OFF to ON is used as the home, without searching for the Z pulse.
25	Similar to mode 9. During reverse operation, the position where the HSW status changes from OFF to ON is used as the home, without searching for the Z pulse.
26	Similar to mode 10. During forward operation, the position where the HSW status changes from ON to OFF is used as the home, without searching for the Z pulse.
27	Similar to mode 11. During forward operation, the position where the HSW status changes from ON to OFF is used as the home, without searching for the Z pulse.
28	Similar to mode 12. During reverse operation, the position where the HSW status changes from OFF to ON is used as the home, without searching for the Z pulse.
29	Similar to mode 13. During forward operation, the position where the HSW status changes from OFF to ON is used as the home, without searching for the Z pulse.
30	Similar to mode 14. During reverse operation, the position where the HSW status changes from ON to OFF is used as the home, without searching for the Z pulse.
31	Reserved
32	Reserved
33	The motor starts to operate in the reverse direction, and searches for the nearest Z pulse position as the home.
34	The motor starts to operate in the forward direction, and searches for the nearest Z pulse position as the home.
35	The current position is used as the home.

Home mode:

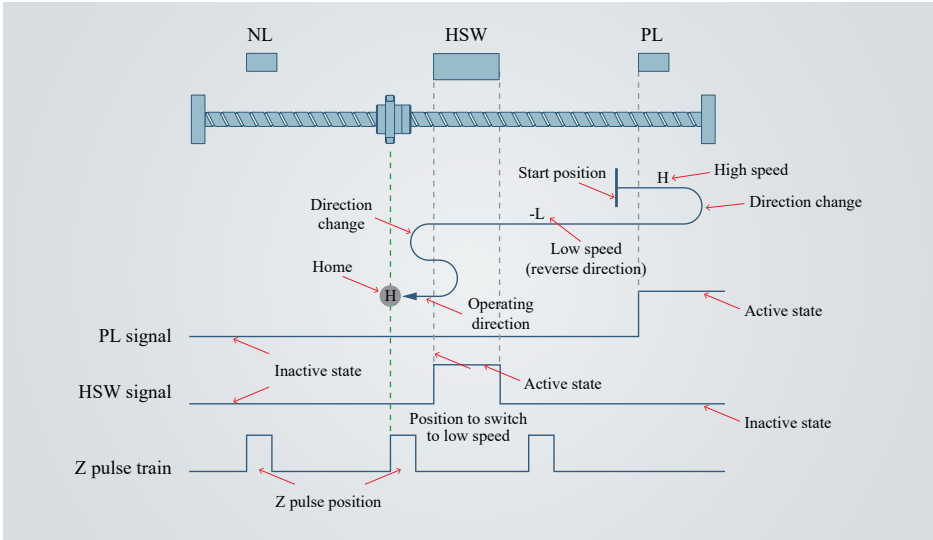


Figure 4-23 Definition of the home mode

Mode-2: The mode is used to search for the mechanical extreme position and Z pulse in the forward direction.

- The motor starts to operate in the forward direction at a high speed. After the motor runs into the mechanical extreme position, if the torque reaches the torque limit value, the speed is near zero, and the state is maintained for a certain period of time, the axis reaches the mechanical extreme position. The motor switches to operate in the reverse direction at a low speed, and searches for the nearest Z pulse position as the home.

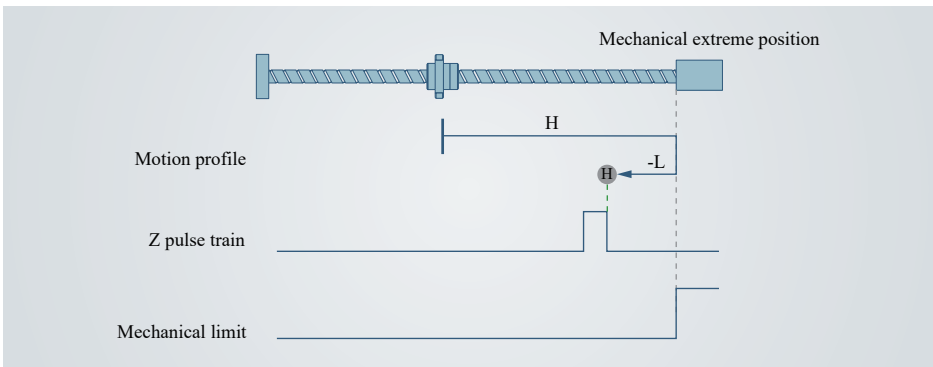


Figure 4-24 Home mode-2 profile and signal status

Mode-1: The mode is used to search for the mechanical extreme position and Z pulse in the reverse direction.

- The motor starts to operate in the reverse direction at a high speed. After the motor runs into the mechanical extreme position, if the torque reaches the torque limit value, the speed is near zero, and the state is maintained for a certain period of time, the axis reaches the mechanical extreme position. The motor switches to operate in the forward direction at a low speed, and searches for the nearest Z pulse position as the home.

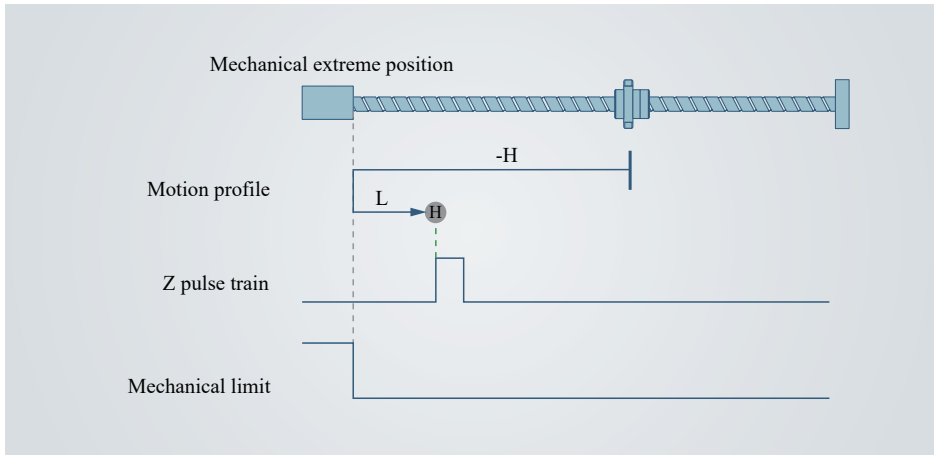


Figure 4-25 Home mode-1 profile and signal status

Mode 1: Search for the NL and Z pulse.

- If the NL is inactive upon startup, the motor operates in the reverse direction at a high speed. When the NL status changes from OFF to ON, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the NL status changes from ON to OFF during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.
- If the NL is active upon startup, the motor operates in the forward direction at a low speed. When the NL status changes from ON to OFF during forward operation, the motor continues the forward operation to search for the nearest Z pulse position as the home.

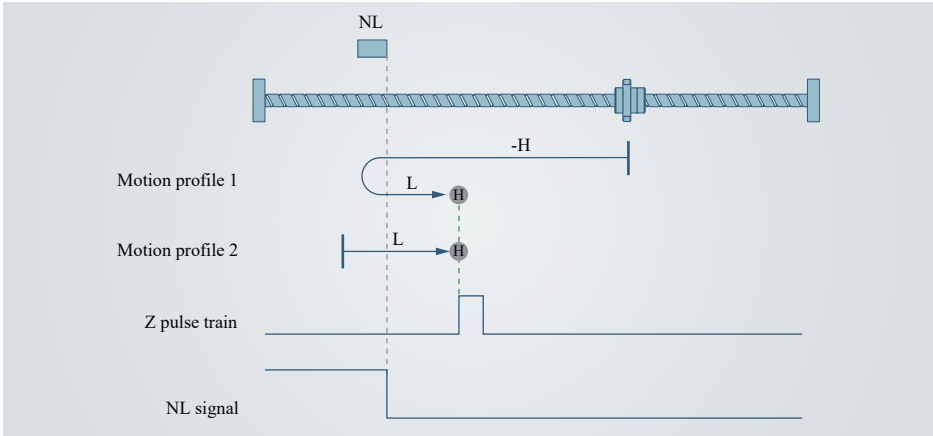


Figure 4-26 Home mode 1 profile and signal status

Mode 2: Search for the PL and Z pulse.

- If the PL is inactive upon startup, the motor operates in the forward direction at a high speed. When the PL status changes from OFF to ON, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the PL status changes from ON to OFF during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.
- If the PL is active upon startup, the motor operates in the reverse direction at a low speed. When the PL status changes from ON to OFF during reverse operation, the motor continues the reverse operation to search for the nearest Z pulse position as the home.

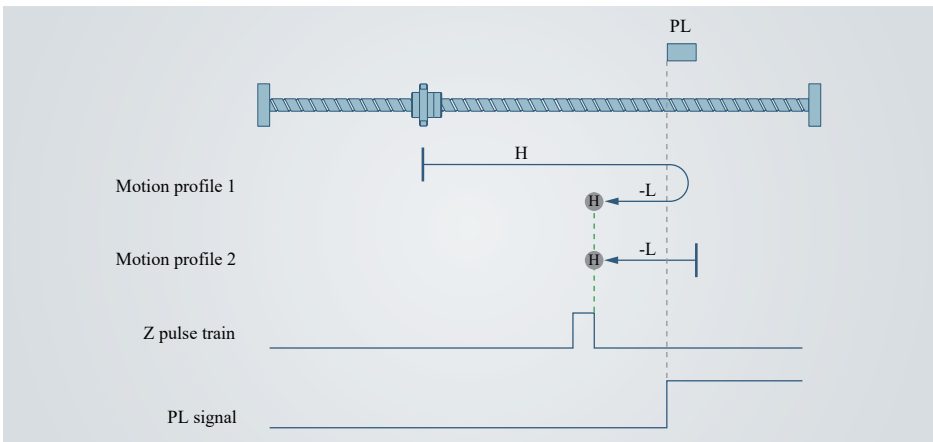


Figure 4-27 Home mode 2 profile and signal status

Mode 3: When the HSW status changes from ON to OFF during reverse operation, the motor searches for the position and Z pulse.

- If the HSW is inactive upon startup, the motor operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.
- If the HSW is active upon startup, the motor operates in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation at a low speed, the motor searches for the nearest Z pulse position as the home.
- In this mode, no matter whether the NL or PL is in ON state, the homing process is stopped and an alarm is reported.

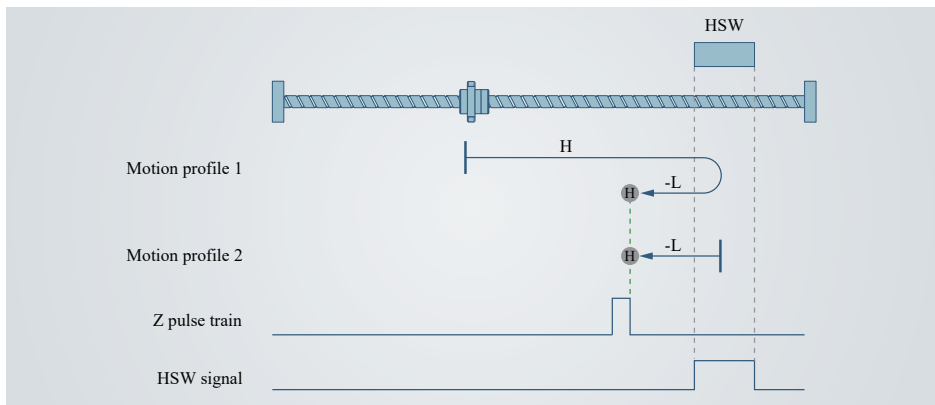


Figure 4-28 Home mode 3 profile and signal status

Mode 4: When the HSW status changes from OFF to ON during forward operation, the motor searches for the position and Z pulse.

- If the HSW is inactive upon startup, the motor operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates to stop, returns to the position where the HSW is inactive at a low speed, decelerates to stop again, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.
- If the HSW is active upon startup, the motor operates in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.
- In this mode, no matter whether the NL or PL is in ON state, the homing process is stopped and an alarm is reported.

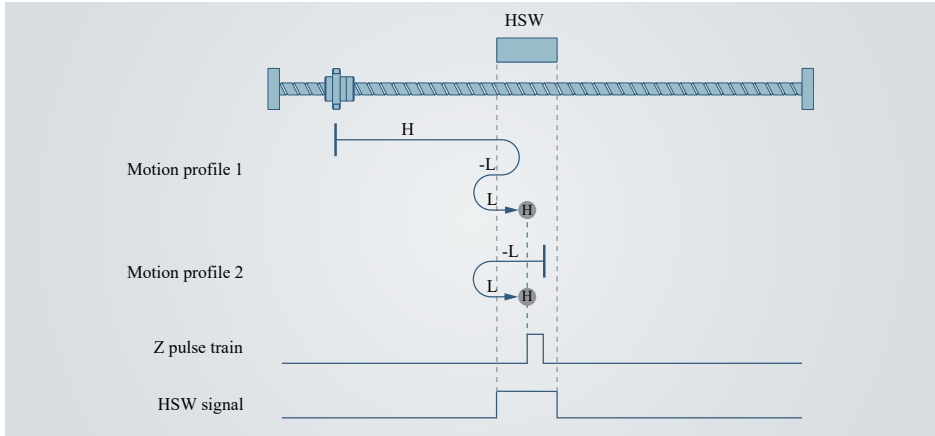


Figure 4-29 Home mode 4 profile and signal status

Mode 5: When the HSW status changes from ON to OFF during forward operation, the motor searches for the position and Z pulse.

- If the HSW is inactive upon startup, the motor operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.
- If the HSW is active upon startup, the motor operates in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor searches for the nearest Z pulse position as the home.
- In this mode, no matter whether the NL or PL is in ON state, the homing process is stopped and an alarm is reported.

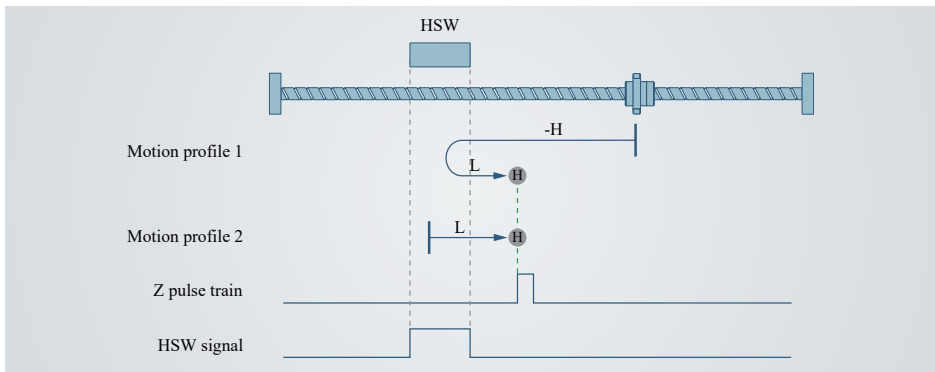


Figure 4-30 Home mode 5 profile and signal status

Mode 6: When the HSW status changes from OFF to ON during reverse operation, the motor searches for the position and Z pulse.

- If the HSW is inactive upon startup, the motor operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates to stop, returns to the position where the HSW is inactive at a low speed, decelerates to stop again, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.
- If the HSW is active upon startup, the motor operates in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.
- In this mode, no matter whether the NL or PL is in ON state, the homing process is stopped and an alarm is reported.

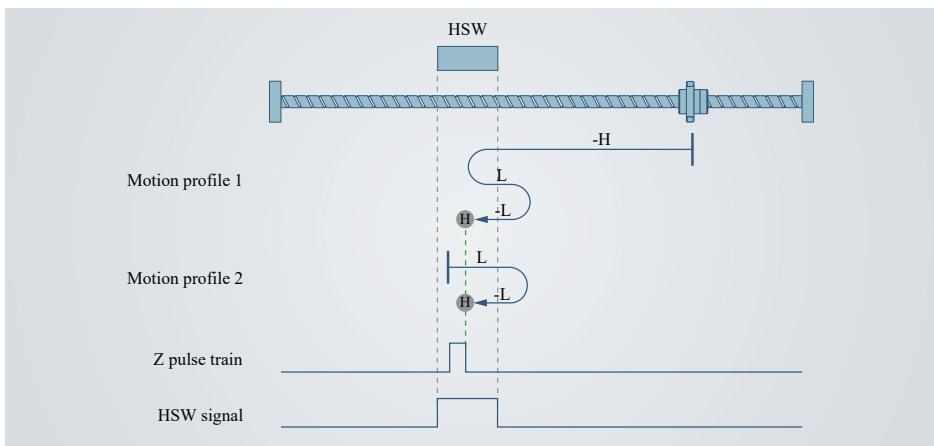


Figure 4-31 Home mode 6 profile and signal status

Mode 7: When the HSW status changes from ON to OFF during reverse operation, the motor searches for the position and Z pulse and automatically turns to another direction upon the PL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the PL is in ON state, the motor decelerates to stop, and then operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates and continues to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, returns to the position where the HSW is active at a low speed, decelerates to stop again (if the active HSW range is narrow, it may enter the inactive HSW position range on the other side), and switches to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse

operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.

- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.
- If the HSW is active upon startup, the motor operates in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation at a low speed, the motor searches for the nearest Z pulse position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the PL for the first time during forward operation. When the motor encounters the ON state of the NL or encounters the ON state of the PL again, the homing process is stopped and an alarm is reported.

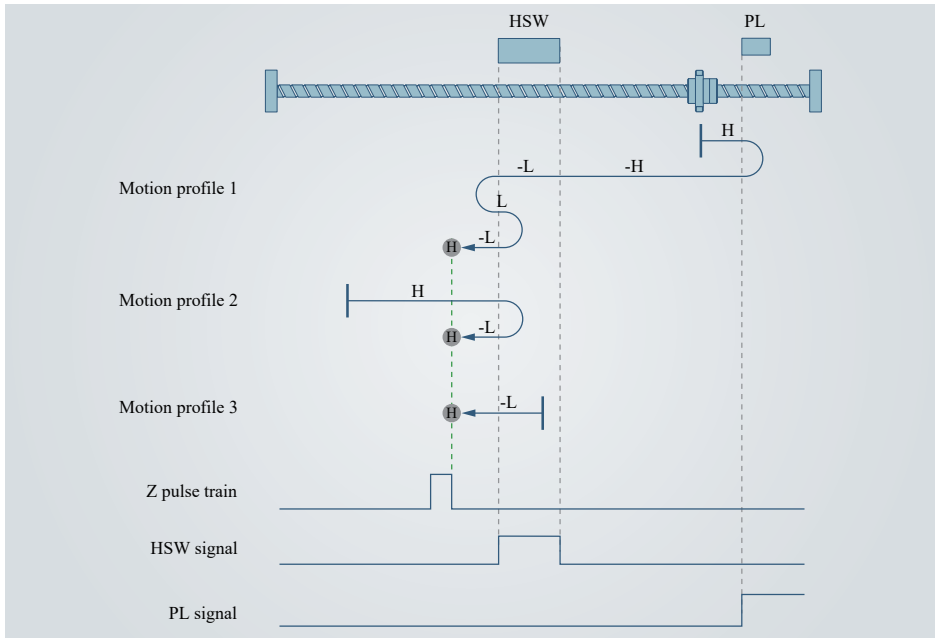


Figure 4-32 Home mode 7 profile and signal status

Mode 8: When the HSW status changes from OFF to ON during forward operation, the motor searches for the position and Z pulse and automatically turns to another direction upon the PL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the PL is in ON state, the motor decelerates to stop, and then operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON

during reverse operation, the motor decelerates, and continues to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.

- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates to stop, returns to the position where the HSW is inactive at a low speed, decelerates to stop again, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.
- If the HSW is active upon startup, the motor operates in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the PL for the first time during forward operation. When the motor encounters the ON state of the NL or encounters the ON state of the PL again, the homing process is stopped and an alarm is reported.

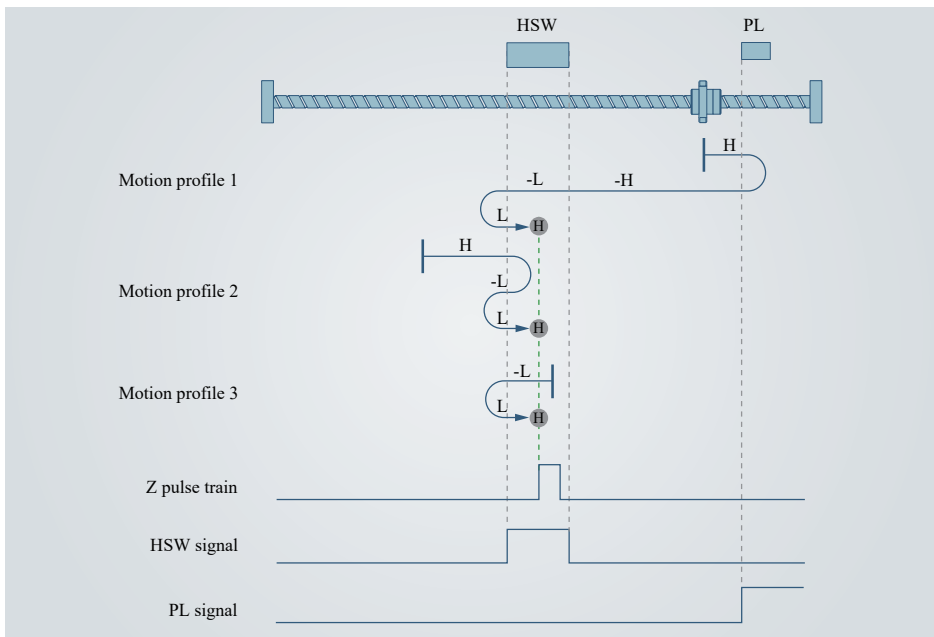


Figure 4-33 Home mode 8 profile and signal status

Mode 9: When the HSW status changes from OFF to ON during reverse operation, the motor searches for the position and Z pulse and automatically turns to another direction upon the PL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the PL is in ON state, the motor decelerates to stop, and then operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates to stop, returns to the position where the HSW is inactive at a low speed, decelerates to stop again, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.
- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates, and continues to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.
- If the HSW is active upon startup, the motor operates in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the PL for the first time during forward operation. When the motor encounters the ON state of the NL or encounters the ON state of the PL again, the homing process is stopped and an alarm is reported.

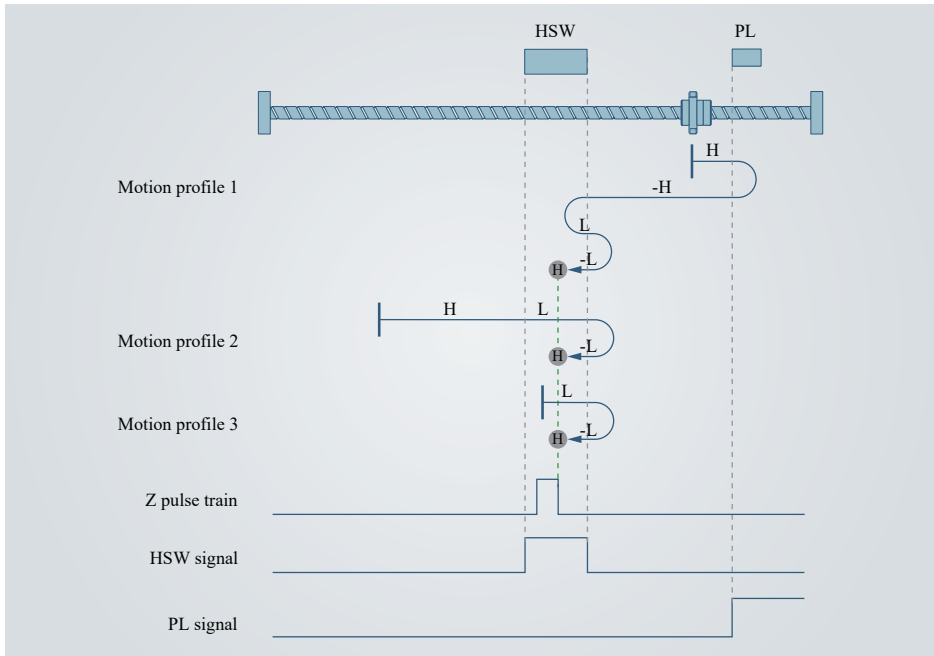


Figure 4-34 Home mode 9 profile and signal status

Mode 10: When the HSW status changes from ON to OFF during forward operation, the motor searches for the position and Z pulse and automatically turns to another direction upon the PL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the PL is in ON state, the motor decelerates to stop, and then operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.
- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates, and continues to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, returns to the position where the HSW is active at a low speed, decelerates to stop again (if the active HSW range is narrow, it may enter the inactive HSW position range on the other side), and switches to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.

- If the HSW is active upon startup, the motor operates in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor searches for the nearest Z pulse position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the PL for the first time during forward operation. When the motor encounters the ON state of the NL or encounters the ON state of the PL again, the homing process is stopped and an alarm is reported.

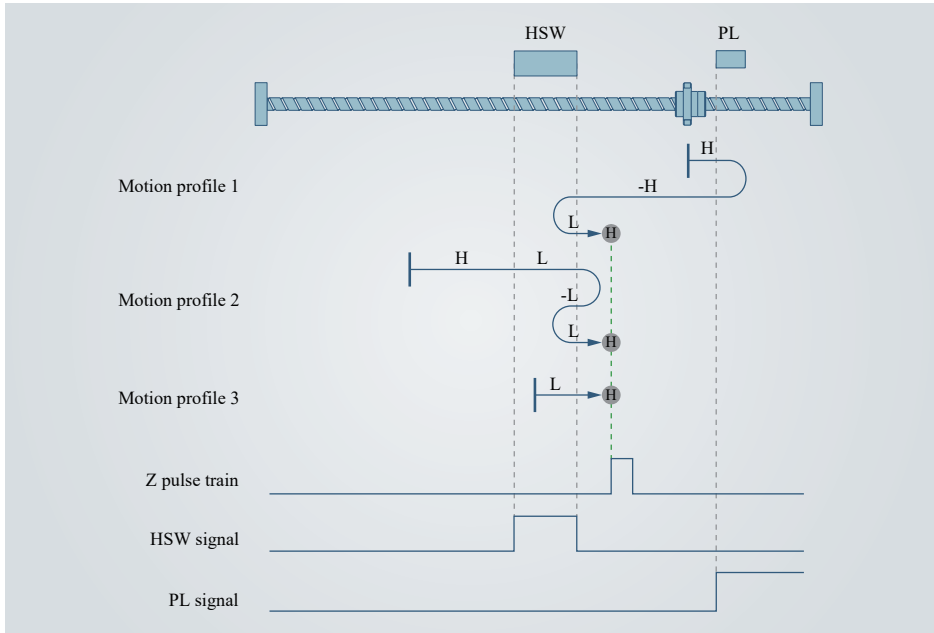


Figure 4-35 Home mode 10 profile and signal status

Mode 11: When the HSW status changes from ON to OFF during forward operation, the motor searches for the position and Z pulse and automatically turns to the forward direction upon the NL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.
- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the NL is in ON state, the motor decelerates to stop, and then operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates and continues to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, returns to

the position where the HSW is active at a low speed, decelerates to stop again (if the active HSW range is narrow, it may enter the inactive HSW position range on the other side), and switches to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.

- If the HSW is active upon startup, the motor operates in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor searches for the nearest Z pulse position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the NL for the first time during reverse operation. When the motor encounters the ON state of the PL or encounters the ON state of the NL again, the homing process is stopped and an alarm is reported.

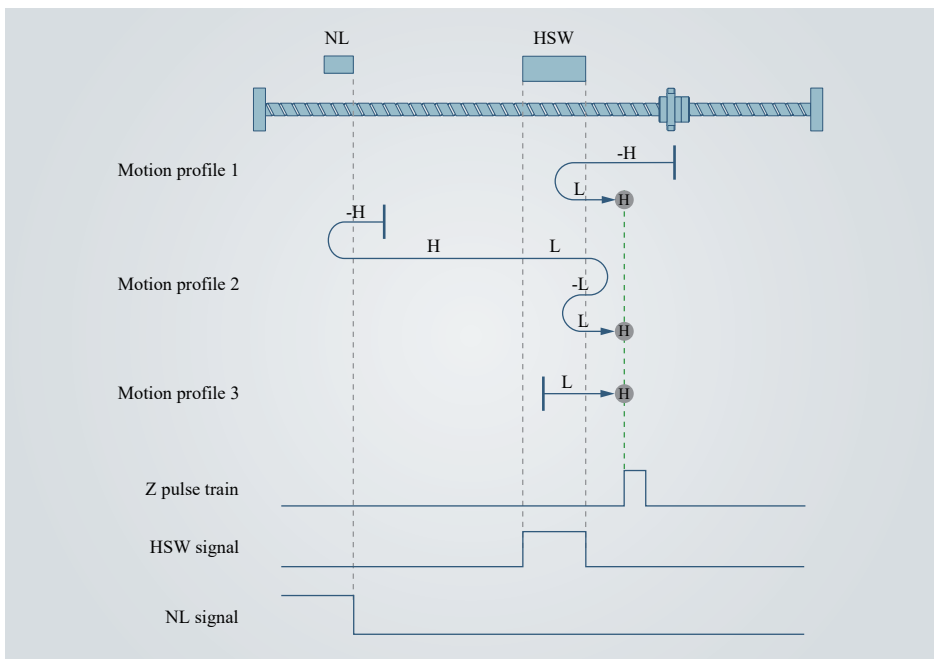


Figure 4-36 Home mode 11 profile and signal status

Mode 12: When the HSW status changes from OFF to ON during reverse operation, the motor searches for the position and Z pulse and automatically turns to another direction upon the NL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates to stop, returns to the position where the HSW is inactive at a low speed, decelerates to stop again, and then switches to operate in the reverse direction at a low speed.

When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.

- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the NL is in ON state, the motor decelerates to stop, and then operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates, and continues to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.
- If the HSW is active upon startup, the motor operates in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the NL for the first time during reverse operation. When the motor encounters the ON state of the PL or encounters the ON state of the NL again, the homing process is stopped and an alarm is reported.

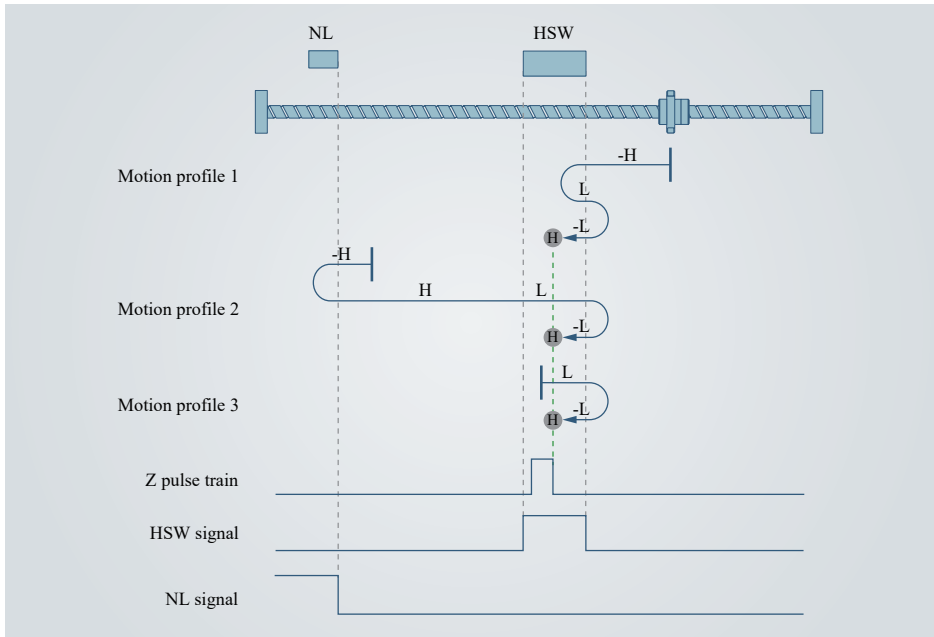


Figure 4-37 Home mode 12 profile and signal status

Mode 13: When the HSW status changes from OFF to ON during forward operation, the motor searches for the position and Z pulse and automatically turns to another direction upon the NL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates, and continues to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.
- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the NL is in ON state, the motor decelerates to stop, and then operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates to stop, returns to the position where the HSW is inactive at a low speed, decelerates to stop again, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.
- If the HSW is active upon startup, the motor operates in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor continues the forward operation to search for the nearest Z pulse position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the NL for the first time during reverse operation. When the motor encounters the ON state of the PL or encounters the ON state of the NL again, the homing process is stopped and an alarm is reported.

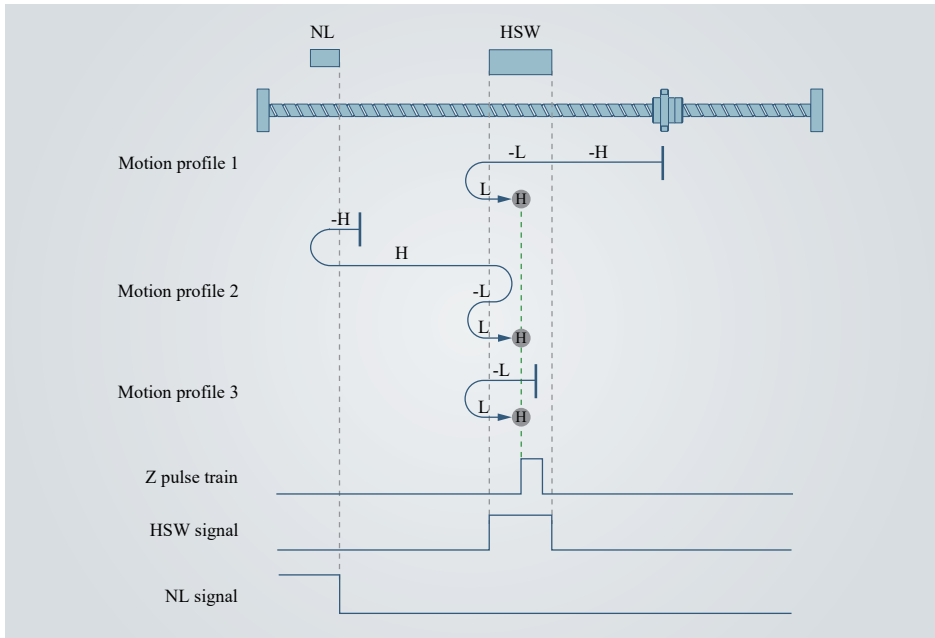


Figure 4-38 Home mode 13 profile and signal status

Mode 14: When the HSW status changes from ON to OFF during reverse operation, the motor searches for the position and Z pulse and automatically turns to another direction upon the NL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates and continues to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, returns to the position where the HSW is active at a low speed, decelerates to stop again (if the active HSW range is narrow, it may enter the inactive HSW position range on the other side), and switches to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.
- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the NL is in ON state, the motor decelerates to stop, and then operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation at a low speed, the motor continues the reverse operation to search for the nearest Z pulse position as the home.

- If the HSW is active upon startup, the motor operates in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation at a low speed, the motor searches for the nearest Z pulse position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the NL for the first time during reverse operation. When the motor encounters the ON state of the PL or encounters the ON state of the NL again, the homing process is stopped and an alarm is reported.

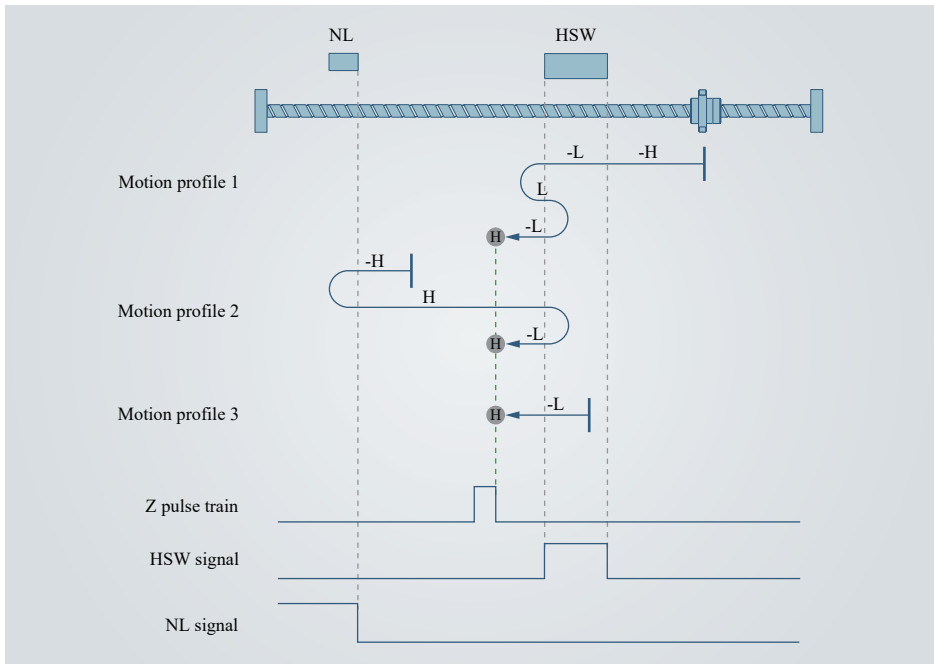


Figure 4-39 Home mode 14 profile and signal status

Mode 15 and mode 16: Reserved

Mode 17: Search for the NL.

- If the NL is inactive upon startup, the motor operates in the reverse direction at a high speed. When the NL status changes from OFF to ON, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the NL status changes from ON to OFF during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the NL is active upon startup, the motor operates in the forward direction at a low speed. When the NL status changes from ON to OFF during forward operation, the motor decelerates to stop and uses the stop position as the home.

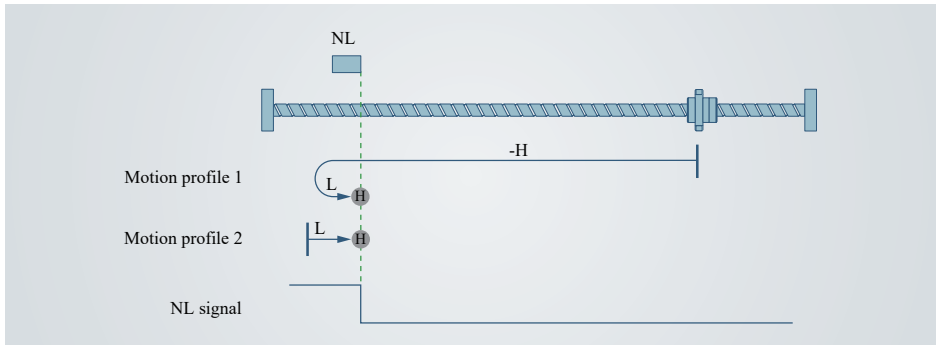


Figure 4-40 Home mode 17 profile and signal status

Mode 18: Search for the PL.

- If the PL is inactive upon startup, the motor operates in the forward direction at a high speed. When the PL status changes from OFF to ON, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the PL status changes from ON to OFF during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the PL is active upon startup, the motor operates in the reverse direction at a low speed. When the PL status changes from ON to OFF during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.

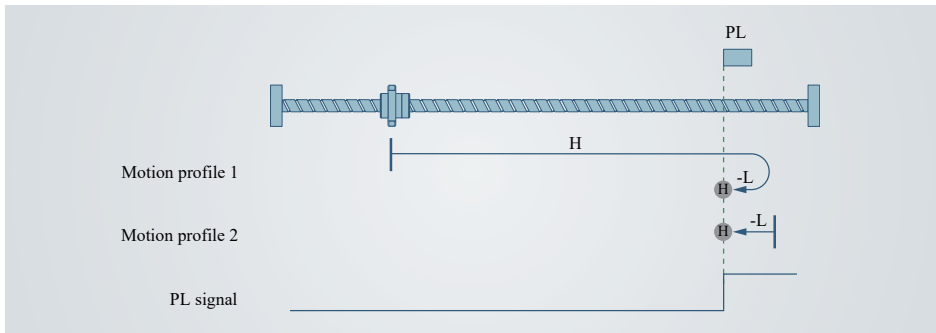


Figure 4-41 Home mode 18 profile and signal status

Mode 19: When the HSW status changes from ON to OFF during reverse operation, the motor searches for the position.

- If the HSW is inactive upon startup, the motor operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from

ON to OFF during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.

- If the HSW is active upon startup, the motor operates in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation, the motor uses the stop position as the home.
- In this mode, no matter whether the NL or PL is in ON state, the homing process is stopped and an alarm is reported.

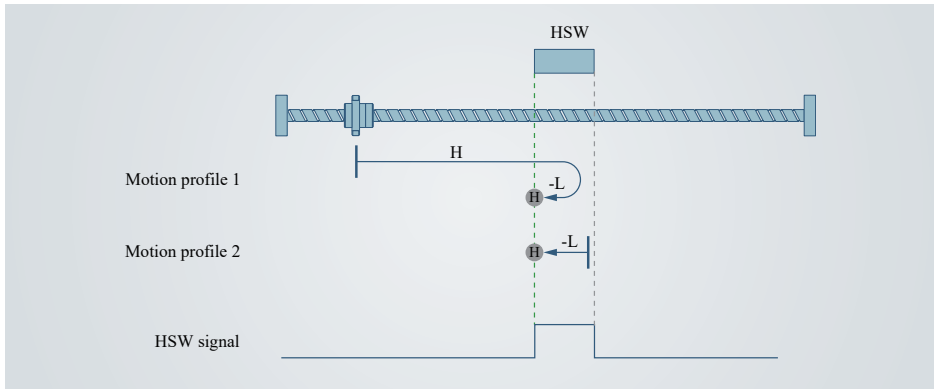


Figure 4-42 Home mode 19 profile and signal status

Mode 20: When the HSW status changes from OFF to ON during forward operation, the motor searches for the position.

- If the HSW is inactive upon startup, the motor operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates to stop, returns to the position where the HSW is inactive at a low speed, decelerates to stop again, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is active upon startup, the motor operates in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- In this mode, no matter whether the NL or PL is in ON state, the homing process is stopped and an alarm is reported.

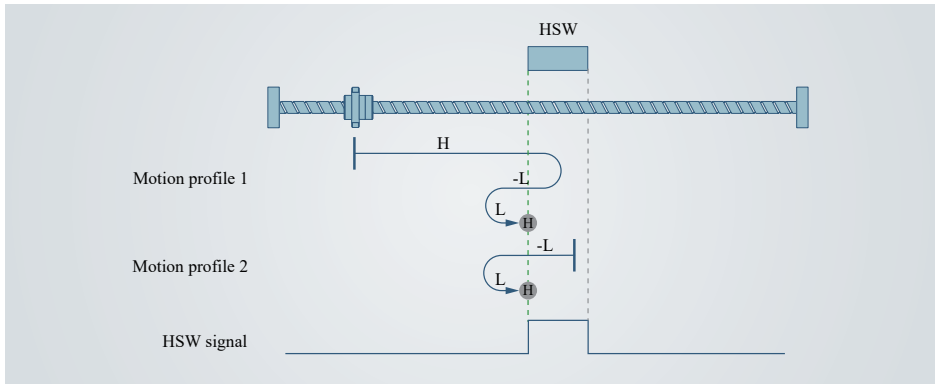


Figure 4-43 Home mode 20 profile and signal status

Mode 21: When the HSW status changes from ON to OFF during forward operation, the motor searches for the position.

- If the HSW is inactive upon startup, the motor operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is active upon startup, the motor operates in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation, the motor uses the stop position as the home.
- In this mode, no matter whether the NL or PL is in ON state, the homing process is stopped and an alarm is reported.

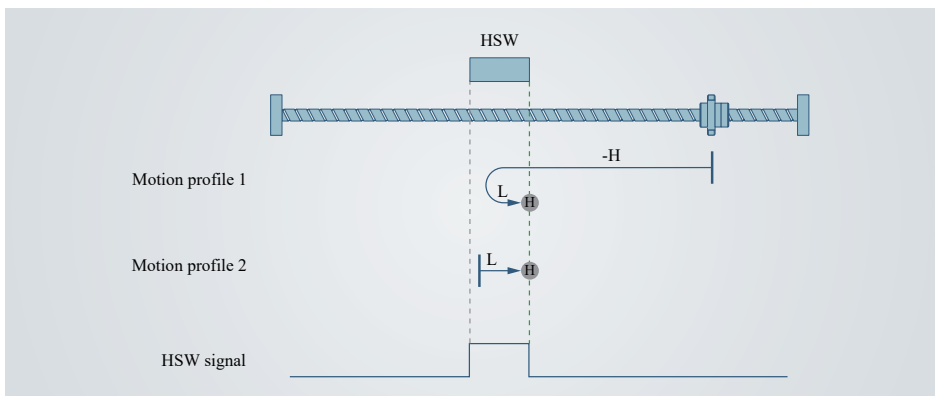


Figure 4-44 Home mode 21 profile and signal status

Mode 22: Search for the PL.

- If the HSW is inactive upon startup, the motor operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates to stop, returns to the position where the HSW is inactive at a low speed, decelerates to stop again, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is active upon startup, the motor operates in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- In this mode, no matter whether the NL or PL is in ON state, the homing process is stopped and an alarm is reported.

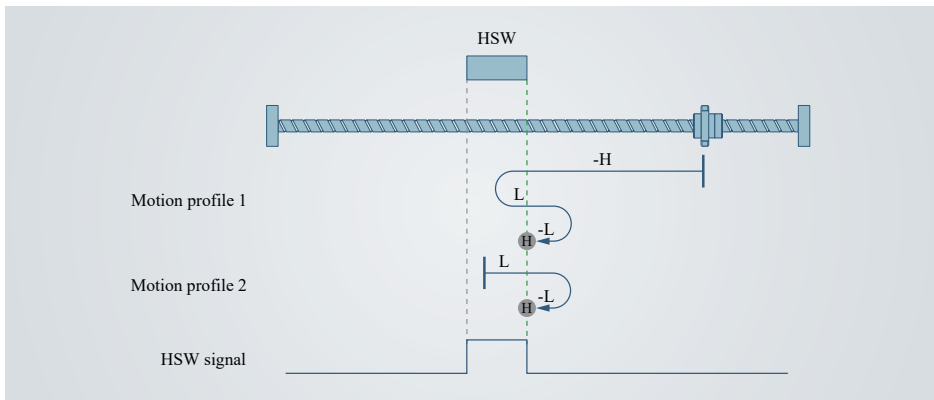


Figure 4-45 Home mode 22 profile and signal status

Mode 23: When the HSW status changes from ON to OFF during reverse operation, the motor searches for the position and automatically turns to another direction upon the PL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the PL is in ON state, the motor decelerates to stop, and then operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates and continues to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, returns to the position where the HSW is active at a low speed, decelerates to stop again (if the active HSW range is narrow, it may enter the inactive HSW position range on the other side), and switches to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates

in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.

- If the HSW is active upon startup, the motor operates in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation, the motor uses the stop position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the PL for the first time during forward operation. When the motor encounters the ON state of the NL or encounters the ON state of the PL again, the homing process is stopped and an alarm is reported.

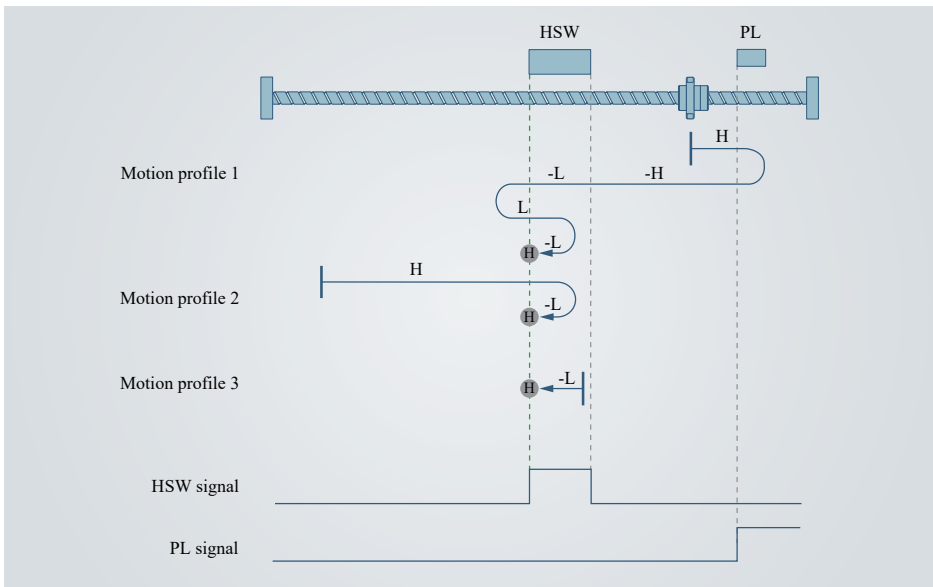


Figure 4-46 Home mode 23 profile and signal status

Mode 24: When the HSW status changes from OFF to ON during forward operation, the motor searches for the position and automatically turns to another direction upon the PL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the PL is in ON state, the motor decelerates to stop, and then operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates, and continues to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor decelerates to stop and uses the stop position

as the home.

- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates to stop, returns to the position where the HSW is inactive at a low speed, decelerates to stop again, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is active upon startup, the motor operates in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the PL for the first time during forward operation. When the motor encounters the ON state of the NL or encounters the ON state of the PL again, the homing process is stopped and an alarm is reported.

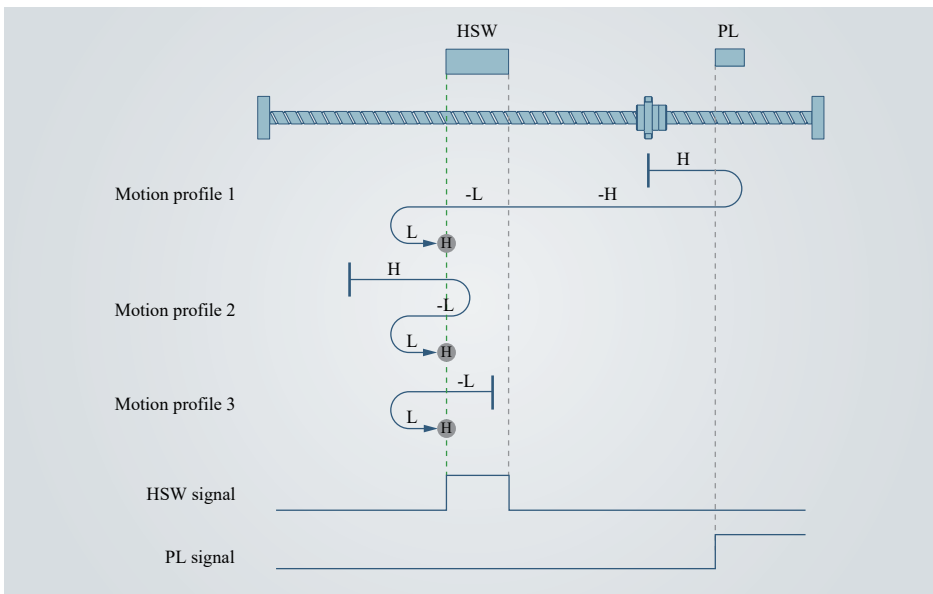


Figure 4-47 Home mode 24 profile and signal status

Mode 25: When the HSW status changes from OFF to ON during reverse operation, the motor searches for the position and automatically turns to another direction upon the PL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the PL is in ON state, the motor decelerates to stop,

and then operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates to stop, returns to the position where the HSW is inactive at a low speed, decelerates to stop again, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.

- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates, and continues to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is active upon startup, the motor operates in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the PL for the first time during forward operation. When the motor encounters the ON state of the NL or encounters the ON state of the PL again, the homing process is stopped and an alarm is reported.

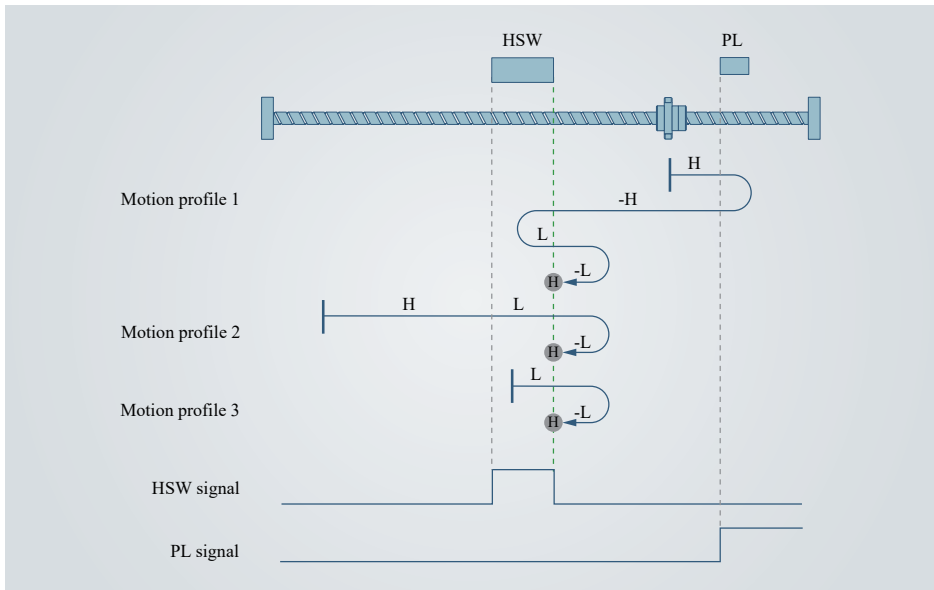


Figure 4-48 Home mode 25 profile and signal status

Mode 26: When the HSW status changes from ON to OFF during forward operation, the motor searches for the position and automatically turns to another direction upon the PL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the PL is in ON state, the motor decelerates to stop, and then operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates and continues to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, returns to the position where the HSW is active at a low speed, decelerates to stop again (if the active HSW range is narrow, it may enter the inactive HSW position range on the other side), and switches to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is active upon startup, the motor operates in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation, the motor uses the stop position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the PL for the first time during forward operation. When the motor encounters the ON state of the NL or encounters the ON state of the PL again, the homing process is stopped and an alarm is reported.

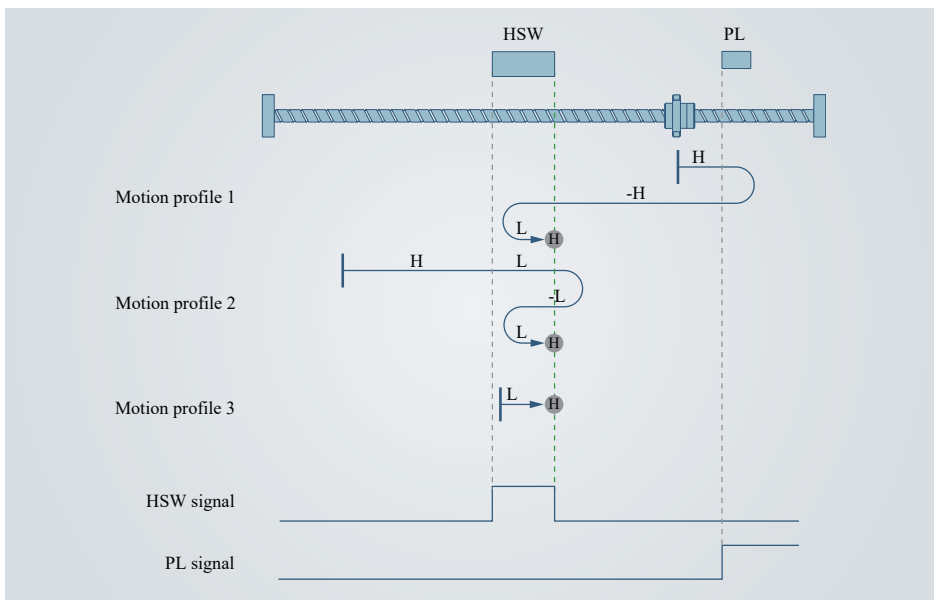


Figure 4-49 Home mode 26 profile and signal status

Mode 27: When the HSW status changes from ON to OFF during forward operation, the motor searches for the position and automatically turns to another direction upon the NL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the NL is in ON state, the motor decelerates to stop, and then operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates and continues to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, returns to the position where the HSW is active at a low speed, decelerates to stop again (if the active HSW range is narrow, it may enter the inactive HSW position range on the other side), and switches to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is active upon startup, the motor operates in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation, the motor uses the stop position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the NL for the first time during reverse operation. When the motor encounters the ON state of the PL or encounters the ON state of the NL again, the homing process is stopped and an alarm is reported.

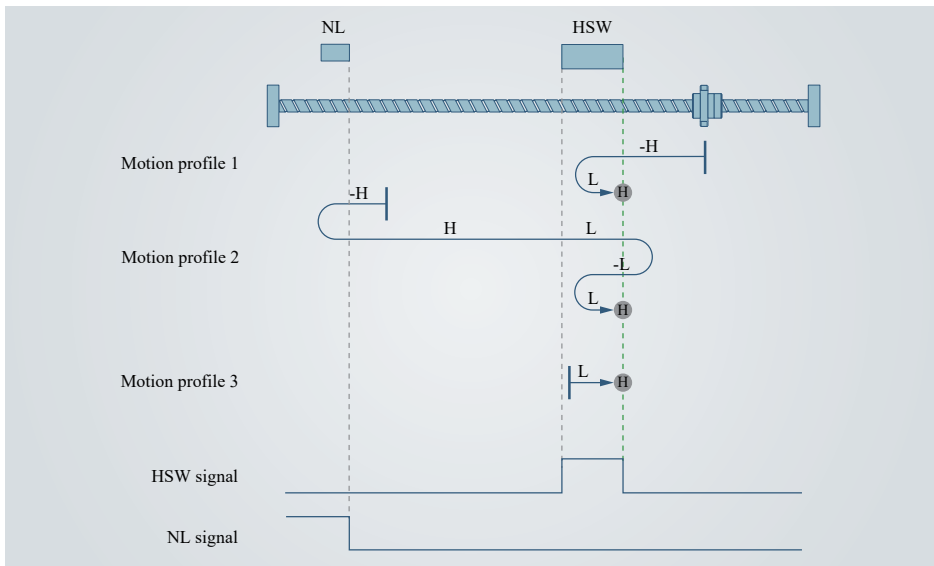


Figure 4-50 Home mode 27 profile and signal status

Mode 28: When the HSW status changes from OFF to ON during reverse operation, the motor searches for the position and automatically turns to another direction upon the NL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates to stop, returns to the position where the HSW is inactive at a low speed, decelerates to stop again, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the NL is in ON state, the motor decelerates to stop, and then operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates, and continues to operate in the forward direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is active upon startup, the motor operates in the forward direction at a low speed. When the HSW status changes from ON to OFF during forward operation, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from OFF to ON during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the NL for the first time during reverse operation. When the motor encounters the ON state of the PL or encounters the ON state of the NL again, the homing process is stopped and an alarm is reported.

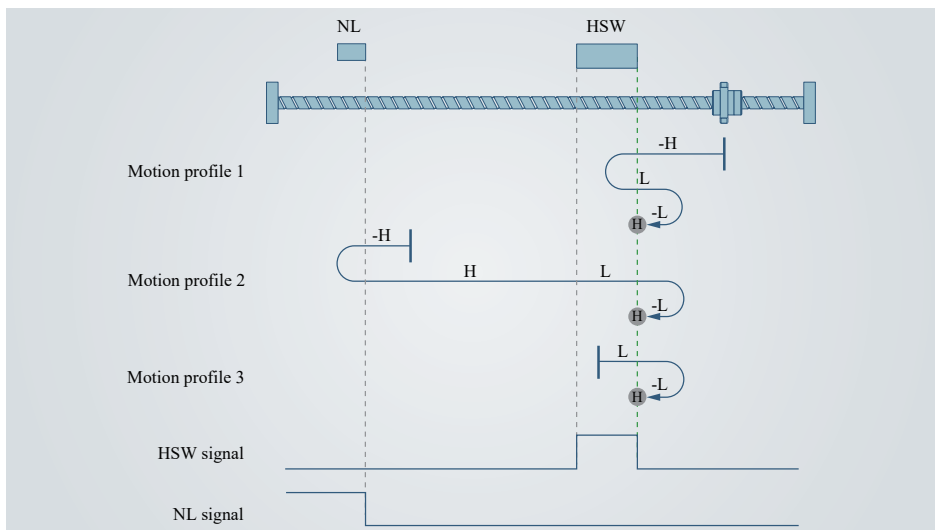


Figure 4-51 Home mode 28 profile and signal status

Mode 29: When the HSW status changes from OFF to ON during forward operation, the motor searches for the position and automatically turns to another direction upon the NL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates, and continues to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the NL is in ON state, the motor decelerates to stop, and then operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates to stop, returns to the position where the HSW is inactive at a low speed, decelerates to stop again, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is active upon startup, the motor operates in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation, the motor decelerates to stop, and then switches to operate in the forward direction at a low speed. When the HSW status changes from OFF to ON during forward operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the NL for the first time during reverse operation. When the motor encounters the ON state of the PL or encounters the ON state of the NL again, the homing process is stopped and an alarm is reported.

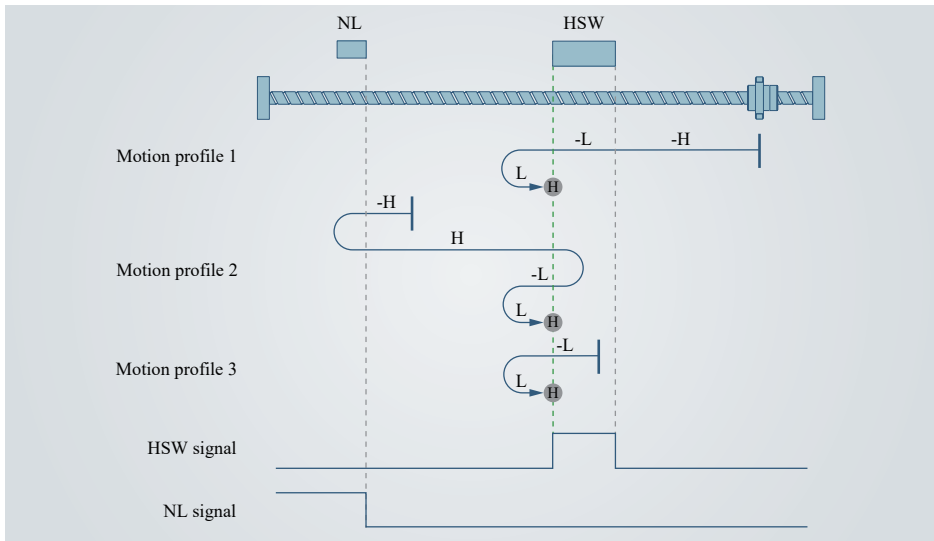


Figure 4-52 Home mode 29 profile and signal status

Mode 30: When the HSW status changes from ON to OFF during reverse operation, the motor searches for the position and automatically turns to another direction upon the NL.

- If the HSW is inactive and located in the forward direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the HSW status changes from OFF to ON during reverse operation, the motor decelerates and continues to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF, the motor decelerates to stop, returns to the position where the HSW is active at a low speed, decelerates to stop again (if the active HSW range is narrow, it may enter the inactive HSW position range on the other side), and switches to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is inactive and located in the reverse direction of the HSW upon startup, the motor operates in the reverse direction at a high speed. When the NL is in ON state, the motor decelerates to stop, and then operates in the forward direction at a high speed. When the HSW status changes from OFF to ON during forward operation, the motor decelerates to stop, and then switches to operate in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation at a low speed, the motor decelerates to stop and uses the stop position as the home.
- If the HSW is active upon startup, the motor operates in the reverse direction at a low speed. When the HSW status changes from ON to OFF during reverse operation, the motor uses the stop position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the NL for the first time during reverse operation. When the motor encounters the ON state of the PL or encounters the ON state of the NL again, the homing process is stopped and an alarm is reported.

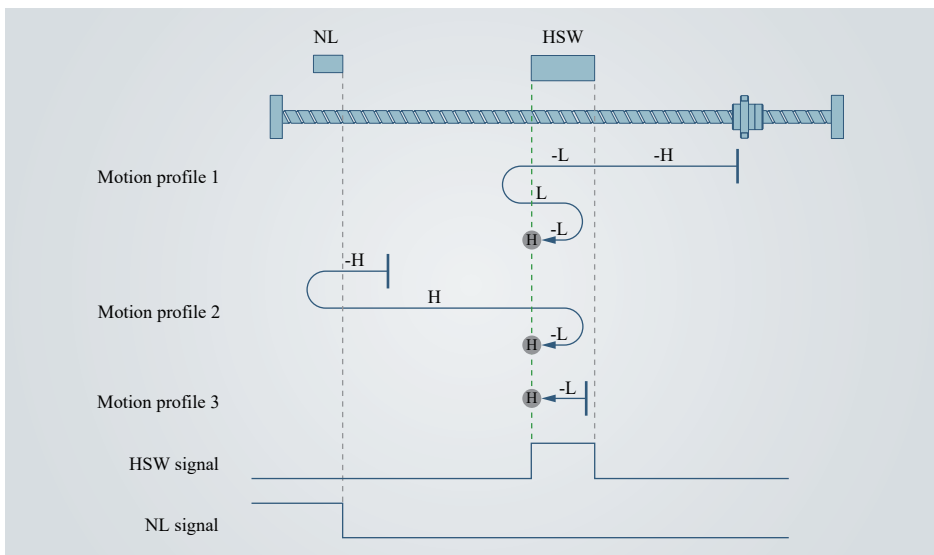


Figure 4-53 Home mode 30 profile and signal status

Mode 31 and mode 32: Reserved

Mode 33: The motor searches for the nearest Z pulse during reverse operation.

- The motor starts to operate in the reverse direction at a low speed, and searches for the nearest Z pulse position as the home. If the motor encounters the ON state of NL before the Z pulse during reverse operation, the motor decelerates to stop, and then operates in the forward direction to search for the nearest Z pulse position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the NL for the first time during reverse operation. When the motor encounters the ON state of the PL or encounters the ON state of the NL again, the homing process is stopped and an alarm is reported.

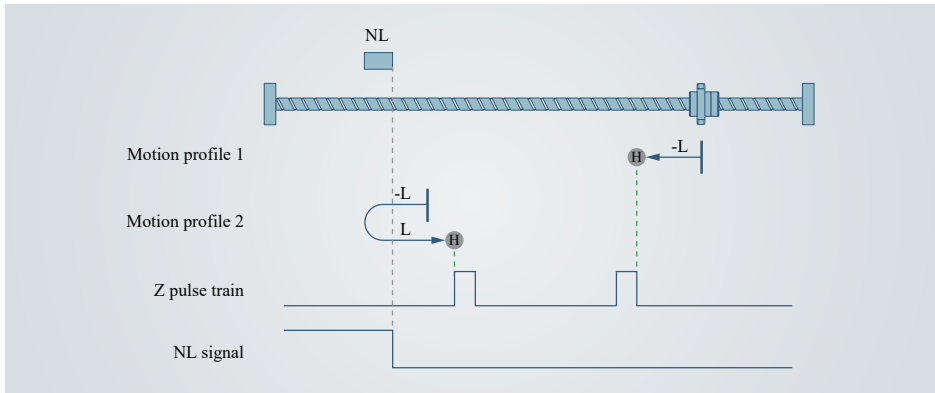


Figure 4-54 Home mode 33 profile and signal status

Mode 34: The motor searches for the nearest Z pulse during forward operation.

- The motor starts to operate in the forward direction at a low speed, and searches for the nearest Z pulse position as the home. If the motor encounters the ON state of PL before the Z pulse during forward operation, the motor decelerates to stop, and then operates in the reverse direction to search for the nearest Z pulse position as the home.
- In this mode, the motor automatically turns to another direction when it encounters the ON state of the PL for the first time during forward operation. When the motor encounters the ON state of the NL or encounters the ON state of the PL again, the homing process is stopped and an alarm is reported.

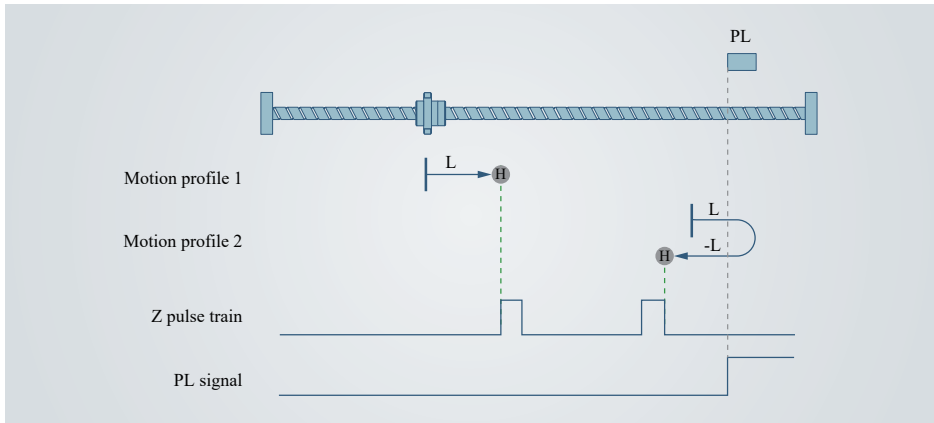


Figure 4-55 Home mode 34 profile and signal status

Mode 35: The current position is used as the home.

- Homing mode 35: The present position is taken as the mechanical home. After homing is triggered (control word 6040h: 0x0F → 0x1F):
 60E6 = 0 (absolute homing):
 6064 (position feedback) is set to 607C (home offset) after homing is done.
 60E6 = 1 (relative homing):
 6064 (position feedback) is the sum of the original value plus 607C (home offset) after homing is done.

4.2 Speed control mode

Set C00.00 to 1 on the servo drive keypad or Synland drive debugging platform to enable the speed control mode. Set the servo drive parameters based on the mechanical structure and specifications. The following part uses the basic parameter setting to describe the speed control mode.

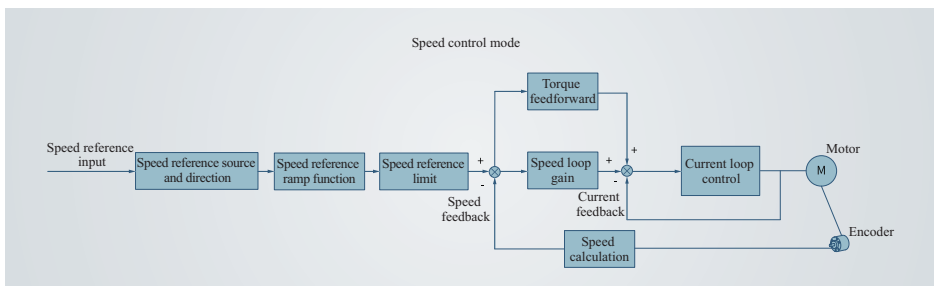


Figure 4-56 Block diagram of speed control